

Fourier Series

Assignment problems:

Half Range Series.

- 1) Find a half range sine series which represents $f(x) = \sin px$, for p is not an integer in the interval $0 < x < \pi$.

Sol: Half range sine series $\Rightarrow a_0 = a_n = 0$

$$\begin{aligned}
 b_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} \sin px \sin nx \, dx \\
 &= \frac{2}{\pi} \left(\frac{1}{2} \right) \int_0^{\pi} \cos(p-n)x - \cos(p+n)x \, dx \\
 &= \frac{1}{\pi} \left[\frac{\sin(p-n)x}{p-n} - \frac{\sin(p+n)x}{p+n} \right]_0^{\pi} \\
 &= \frac{1}{\pi} \left[\frac{-\sin(n-p)\pi}{p-n} - \frac{\sin(n+p)\pi}{p+n} \right] \\
 &= \frac{1}{\pi} \left[\frac{-\sin(n\pi - p\pi)}{p-n} - \frac{\sin(n\pi + p\pi)}{p+n} \right] \\
 &= \frac{1}{\pi} \left[\frac{\cos n\pi \sin p\pi}{p-n} - \frac{\cos n\pi \sin p\pi}{p+n} \right] \\
 &= \frac{1}{\pi} (-1)^n \sin p\pi \left[\frac{1}{p-n} - \frac{1}{p+n} \right] \\
 &= \frac{(-1)^n \sin p\pi}{\pi} \left[\frac{p+n - p+n}{p^2 - n^2} \right] = \frac{2n(-1)^n \sin p\pi}{\pi(p^2 - n^2)}
 \end{aligned}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^n 2n \sin p\pi}{\pi(p^2 - n^2)} \sin nx$$

$$f(x) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n n}{p^2 - n^2} \sin p\pi \sin nx$$

2) Find the sine and cosine half range series in $0 < x < \pi$ if $f(x) = e^x$.

Half range sine : $a_0 = a_n = 0$

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} e^x \sin nx \, dx \\ &= \frac{2}{\pi} \left[\frac{e^x}{1+n^2} (\sin nx - n \cos nx) \right]_0^{\pi} \\ &= \frac{2}{\pi(1+n^2)} \left[e^{\pi} (0 - n \cos n\pi) - e^0 (0 - n \cos 0) \right] \\ &= \frac{2}{\pi(1+n^2)} \left[e^{\pi} (-n(-1)^n) - 1(-n) \right] \end{aligned}$$

$$b_n = \frac{2n}{\pi(1+n^2)} \left[1 - (-1)^n e^{\pi} \right]$$

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \\ &= \sum_{n=1}^{\infty} \frac{2n}{\pi(1+n^2)} \left[1 - (-1)^n e^{\pi} \right] \sin nx \end{aligned}$$

$$f(x) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{n}{1+n^2} \left[1 - (-1)^n e^{\pi} \right] \sin nx$$

Half Range cosine : $b_n = 0$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) \, dx = \frac{2}{\pi} \int_0^{\pi} e^x \, dx = \frac{2}{\pi} (e^x)_0^{\pi}$$

$$a_0 = \frac{2}{\pi} (e^{\pi} - 1)$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx = \frac{2}{\pi} \int_0^{\pi} e^x \cos nx \, dx \\ &= \frac{2}{\pi} \left[\frac{e^x}{1+n^2} (\cos nx + n \sin nx) \right]_0^{\pi} \end{aligned}$$

$$= \frac{2}{\pi(1+n^2)} \left[e^{\pi} (\cos n\pi + 0) - e^0 (1 + 0) \right]$$

$$= \frac{2}{\pi(1+n^2)} \left[e^{\pi} (-1)^n - 1 \right]$$

$$f(x) = \frac{2}{\pi} \frac{(e^{\pi} - 1)}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi(1+n^2)} \left[e^{\pi} (-1)^n - 1 \right] \cos nx$$

$$f(x) = \frac{2(e^{\pi} - 1)}{\pi} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{e^{\pi} (-1)^n - 1}{(1+n^2)} \cos nx$$

3) Obtain the cosine series for

$$f(x) = \begin{cases} \cos x, & 0 < x < \pi/2 \\ 0, & \pi/2 < x < \pi \end{cases}$$

Half range cosine $b_n = 0$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi/2} \cos x dx$$
$$= \frac{2}{\pi} [\sin x]_0^{\pi/2} = \frac{2}{\pi} [1 - 0] = \frac{2}{\pi}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$
$$= \frac{2}{\pi} \int_0^{\pi/2} \cos x \cos nx dx$$
$$= \frac{1}{\pi} \int_0^{\pi/2} [\cos(n+1)x + \cos(n-1)x] dx$$
$$= \frac{1}{\pi} \left[\frac{\sin(n+1)x}{n+1} + \frac{\sin(n-1)x}{n-1} \right]_0^{\pi/2}$$
$$= \frac{1}{\pi} \left[\frac{\sin\left(\frac{n\pi}{2} + \frac{\pi}{2}\right)}{n+1} + \frac{\sin\left(\frac{n\pi}{2} - \frac{\pi}{2}\right)}{n-1} \right]$$
$$= \frac{1}{\pi} \left[\frac{\sin\left(\frac{\pi}{2} + \frac{n\pi}{2}\right)}{n+1} - \frac{\sin\left(\frac{\pi}{2} - \frac{n\pi}{2}\right)}{n-1} \right]$$
$$= \frac{1}{\pi} \left[\frac{\cos \frac{n\pi}{2}}{n+1} - \frac{\cos \frac{n\pi}{2}}{n-1} \right]$$
$$= \frac{1}{\pi} \cos\left(\frac{n\pi}{2}\right) \left[\frac{n-1-n-1}{n^2-1} \right]$$
$$= \frac{1}{\pi} \cos\left(\frac{n\pi}{2}\right) \left(\frac{-2}{n^2-1} \right)$$

$$a_1 = \frac{2}{\pi} \int_0^{\pi} f(x) \cos x dx$$
$$= \frac{2}{\pi} \int_0^{\pi/2} \cos x \cos x dx = \frac{2}{\pi} \int_0^{\pi/2} \cos^2 x dx$$
$$= \frac{2}{\pi} \int_0^{\pi/2} \left(\frac{1 + \cos 2x}{2} \right) dx$$
$$= \frac{1}{\pi} \left(x + \frac{\sin 2x}{2} \right)_0^{\pi/2} = \frac{1}{\pi} \left(\frac{\pi}{2} \right) = \frac{1}{2}$$

$$\begin{aligned}
 b_1 &= \frac{2}{\pi} \int_0^{\pi} f(x) \sin x \, dx = \frac{2}{\pi} \int_0^{\pi/2} \cos x \sin x \, dx \\
 &= \frac{2}{\pi} \int_0^{\pi/2} \frac{\sin 2x}{2} \, dx = \frac{1}{\pi} \left[-\frac{\cos 2x}{2} \right]_0^{\pi/2} \\
 &= \frac{-1}{2\pi} [-1-1] = \frac{-1}{2\pi} (-2) = \frac{1}{\pi}
 \end{aligned}$$

$$\begin{aligned}
 f(x) &= \frac{a_0}{2} + a_1 \cos x + b_1 \sin x + \sum_{n=2}^{\infty} a_n \cos nx + b_n \sin nx \\
 &= \frac{2/\pi}{2} + \frac{\cos x}{2} + \frac{\sin x}{\pi} + \sum_{n=2}^{\infty} \frac{-2}{\pi(n^2-1)} \cos\left(\frac{n\pi}{2}\right) \cos nx \\
 &= \frac{1}{\pi} + \frac{\cos x}{2} + \frac{\sin x}{\pi} - \frac{2}{\pi} \sum_{n=2}^{\infty} \frac{1}{(n^2-1)} \cos\left(\frac{n\pi}{2}\right) \cos nx
 \end{aligned}$$

4) Express $f(x) = \begin{cases} \pi x/4, & 0 < x < \pi/2 \\ \pi/4 (\pi - x), & \pi/2 < x < \pi \end{cases}$ as series of cosines of periodicity 2π .

Half range cosine series: $b_n = 0$

$$\begin{aligned}
 a_0 &= \frac{2}{\pi} \int_0^{\pi} f(x) \, dx = \frac{2}{\pi} \left[\int_0^{\pi/2} \frac{\pi x}{4} \, dx + \int_{\pi/2}^{\pi} \frac{\pi}{4} (\pi - x) \, dx \right] \\
 &= \frac{1}{2} \int_0^{\pi/2} x \, dx + \frac{1}{2} \int_{\pi/2}^{\pi} (\pi - x) \, dx \\
 &= \frac{1}{2} \left[\frac{x^2}{2} \right]_0^{\pi/2} + \frac{1}{2} \left[\pi x - \frac{x^2}{2} \right]_{\pi/2}^{\pi} \\
 &= \frac{1}{2} \left[\frac{\pi^2}{8} \right] + \frac{1}{2} \left[\pi^2 - \frac{\pi^2}{2} - \frac{\pi^2}{2} + \frac{\pi^2}{8} \right] \\
 &= \frac{\pi^2}{16} + \frac{\pi^2}{16} = \frac{\pi^2}{8}
 \end{aligned}$$

$$\begin{aligned}
 a_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx \\
 &= \frac{2}{\pi} \left[\int_0^{\pi/2} \frac{\pi}{4} x \cos nx \, dx + \int_{\pi/2}^{\pi} \frac{\pi}{4} (\pi - x) \cos nx \, dx \right] \\
 &= \frac{1}{2} \int_0^{\pi/2} x \cos nx \, dx + \frac{1}{2} \int_{\pi/2}^{\pi} (\pi - x) \cos nx \, dx
 \end{aligned}$$

$$\begin{aligned}
 a_n &= \frac{1}{2} \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi/2} \\
 &\quad + \frac{1}{2} \left[\frac{(\pi-x) \sin nx}{n} - \frac{\cos nx}{n^2} \right]_{\pi/2}^{\pi} \\
 &= \frac{1}{2} \left[\frac{\pi \sin(\frac{n\pi}{2})}{2n} + \frac{\cos(\frac{n\pi}{2})}{n^2} - 0 - \frac{1}{n^2} \right] \\
 &\quad + \frac{1}{2} \left[0 - \frac{(-1)^n}{n^2} - \frac{\pi \sin(\frac{n\pi}{2})}{2n} + \frac{\cos(\frac{n\pi}{2})}{n^2} \right] \\
 &= \frac{1}{2} \left[\frac{2 \cos(\frac{n\pi}{2})}{n^2} - \frac{(-1)^n}{n^2} - \frac{1}{n^2} \right] \\
 &= \frac{1}{2n^2} [2 \cos(\frac{n\pi}{2}) - (-1)^n - 1]
 \end{aligned}$$

$$\begin{aligned}
 f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \\
 &= \frac{\pi^2/8}{2} + \sum_{n=1}^{\infty} \frac{1}{2n^2} [2 \cos(\frac{n\pi}{2}) - (-1)^n - 1] \cos nx
 \end{aligned}$$

$$f(x) = \frac{\pi^2}{16} + \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n^2} [2 \cos(\frac{n\pi}{2}) - (-1)^n - 1] \cos nx$$

5) Find the half range cosine series for $f(x) = (x-1)^2$ in $0 < x < 1$. Hence show that $\pi^2 = 6 \left[1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right]$. $f(x) = (x-1)^2 \rightarrow \textcircled{1}$

Half range cosine: $b_n = 0$

$$a_0 = \frac{2}{l} \int_0^l f(x) dx, \text{ Here } l=1$$

$$= \frac{2}{1} \int_0^1 (x-1)^2 dx = 2 \left[\frac{(x-1)^3}{3} \right]_0^1$$

$$= \frac{2}{3} [0 - (-1)^3] = \frac{2}{3}$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{1} \int_0^1 (x-1)^2 \cos(n\pi x) dx$$

$$= 2 \left[\frac{(x-1)^2 \sin n\pi x}{n\pi} + \frac{2(x-1) \cos n\pi x}{(n\pi)^2} - \frac{2 \sin n\pi x}{(n\pi)^3} \right]_0^1$$

$$= 2 \left[0 + 0 - \frac{2 \sin n\pi}{(n\pi)^3} - 0 - \frac{2(-1)(1)}{(n\pi)^2} + 0 \right]$$

$$= 2 \left[\frac{2}{(n\pi)^2} \right] = \frac{4}{n^2 \pi^2}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$= \frac{2/3}{2} + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi x$$

$$= \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \cos n\pi x \rightarrow (2)$$

Take $x=0$

$$(1) \Rightarrow [f(x)]_{x=0} = (0-1)^2 = 1 \rightarrow (3)$$

$$(2) \Rightarrow [f(x)]_{x=0} = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \cos(0)$$

$$1 = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \rightarrow (4)$$

From (3) & (4)

$$1 = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$\frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\pi^2 = \frac{12}{2} \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] \Rightarrow \pi^2 = 6 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right]$$

$$6) \text{ If } f(x) = \begin{cases} \sin x, & 0 \leq x \leq \pi/4 \\ \cos x, & \pi/4 \leq x \leq \pi/2 \end{cases}$$

Express it as sine series.

$$\text{Here } a_0, a_n = 0 \text{ and } l = \pi/2$$

$$\begin{aligned} b_n &= \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx \\ &= \frac{2}{\pi/2} \int_0^{\pi/2} f(x) \sin\left(\frac{n\pi x}{\pi/2}\right) dx \\ &= \frac{4}{\pi} \left[\int_0^{\pi/4} \sin x \sin(2nx) dx + \int_{\pi/4}^{\pi/2} \cos x \sin(2nx) dx \right] \\ &= \frac{4}{\pi} [I_1 + I_2]. \end{aligned}$$

$$\begin{aligned} I_1 &= \int_0^{\pi/4} \sin x \sin(2nx) dx \\ &= \frac{1}{2} \int_0^{\pi/4} \cos(2n-1)x - \cos(2n+1)x dx \\ &= \frac{1}{2} \left[\frac{\sin(2n-1)x}{2n-1} - \frac{\sin(2n+1)x}{2n+1} \right]_0^{\pi/4} \\ &= \frac{1}{2} \left[\frac{\sin\left(\frac{n\pi}{2} - \frac{\pi}{4}\right)}{2n-1} - \frac{\sin\left(\frac{n\pi}{2} + \frac{\pi}{4}\right)}{2n+1} \right] \end{aligned}$$

$$\sin\left(\frac{n\pi}{2} - \frac{\pi}{4}\right) = \sin \frac{n\pi}{2} \cos \frac{\pi}{4} - \cos \frac{n\pi}{2} \sin \frac{\pi}{4}$$

$$= \frac{1}{\sqrt{2}} \left[\sin\left(\frac{n\pi}{2}\right) - \cos\left(\frac{n\pi}{2}\right) \right]$$

Similarly

$$\sin\left(\frac{n\pi}{2} + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \left[\sin\left(\frac{n\pi}{2}\right) + \cos\left(\frac{n\pi}{2}\right) \right]$$

$$I_1 = \frac{1}{2} \left[\frac{\frac{1}{\sqrt{2}} \left\{ \sin \frac{n\pi}{2} - \cos \frac{n\pi}{2} \right\}}{2n-1} - \frac{\frac{1}{\sqrt{2}} \left\{ \sin \frac{n\pi}{2} + \cos \frac{n\pi}{2} \right\}}{2n+1} \right]$$

$$I_2 = \int_{\pi/4}^{\pi/2} \cos x \sin(2nx) dx = \frac{1}{2} \int_{\pi/4}^{\pi/2} \sin(2n+1)x + \sin(2n-1)x dx$$

$$= \frac{1}{2} \left[\frac{-\cos(2n+1)x}{2n+1} + \frac{-\cos(2n-1)x}{2n-1} \right]_{\pi/4}^{\pi/2}$$

$$= \frac{1}{2} \left[\frac{-\cos\left(\frac{n\pi}{2} + \frac{\pi}{2}\right)}{2n+1} - \frac{\cos\left(n\pi - \frac{\pi}{2}\right)}{2n-1} + \frac{\cos\left(\frac{n\pi}{2} + \frac{\pi}{4}\right)}{2n+1} + \frac{\cos\left(\frac{n\pi}{2} - \frac{\pi}{4}\right)}{2n-1} \right]$$

$$= \frac{1}{2} \left[0 - 0 + \frac{\frac{1}{\sqrt{2}} \left\{ \cos \frac{n\pi}{2} - \sin \frac{n\pi}{2} \right\}}{2n+1} + \frac{\frac{1}{\sqrt{2}} \left\{ \cos \frac{n\pi}{2} + \sin \frac{n\pi}{2} \right\}}{2n-1} \right]$$

$$= \frac{1}{2} \left[\frac{\frac{1}{\sqrt{2}} \left\{ \cos \frac{n\pi}{2} - \sin \frac{n\pi}{2} \right\}}{2n+1} + \frac{\frac{1}{\sqrt{2}} \left\{ \cos \frac{n\pi}{2} + \sin \frac{n\pi}{2} \right\}}{2n-1} \right]$$

$$I_1 + I_2 = \frac{1}{2} \left[\frac{1}{\sqrt{2}} \left\{ \frac{\sin \frac{n\pi}{2}}{2n-1} - \frac{\cos \frac{n\pi}{2}}{2n-1} - \frac{\sin \frac{n\pi}{2}}{2n+1} - \frac{\cos \frac{n\pi}{2}}{2n+1} + \frac{\cos \frac{n\pi}{2}}{2n+1} - \frac{\sin \frac{n\pi}{2}}{2n+1} + \frac{\cos \frac{n\pi}{2}}{2n-1} + \frac{\sin \frac{n\pi}{2}}{2n-1} \right\} \right]$$

$$= \frac{1}{2\sqrt{2}} \left[2 \sin \frac{n\pi}{2} \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right) \right]$$

$$= \frac{1}{\sqrt{2}} \sin \frac{n\pi}{2} \left[\frac{2n+1-2n+1}{4n^2-1} \right] = \frac{1}{\sqrt{2}} \sin \frac{n\pi}{2} \left(\frac{2}{4n^2-1} \right)$$

$$b_n = \frac{4}{\pi} \left[\frac{2}{\sqrt{2}} \sin \frac{n\pi}{2} \left(\frac{1}{4n^2-1} \right) \right] = \frac{8 \sin\left(\frac{n\pi}{2}\right)}{\sqrt{2}\pi(4n^2-1)}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{1}\right) + b_n \sin\left(\frac{n\pi x}{1}\right)$$

$$f(x) = \frac{8}{\sqrt{2}\pi} \sum_{n=1}^{\infty} \frac{1}{(4n^2-1)} \sin\left(\frac{n\pi x}{2}\right) \sin(2nx)$$

Problems on $(-l, l)$ & $(-\pi, \pi)$:

7) Expand $f(x)$ in a fourier series where

$$f(x) = \begin{cases} -\pi, & (-\pi, 0) \\ x, & (0, \pi) \end{cases}$$

Deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{8}$.

$$f(x) = \begin{cases} -\pi, & (-\pi, 0) \\ x, & (0, \pi) \end{cases} \rightarrow \textcircled{1}$$

Neither odd nor even function.

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 -\pi dx + \int_0^{\pi} x dx \right]$$

$$= \frac{1}{\pi} \pi (x)_{-\pi}^0 + \frac{1}{\pi} \left(\frac{x^2}{2} \right)_0^{\pi}$$

$$= -(0 + \pi) + \frac{1}{\pi} \left(\frac{\pi^2}{2} - 0 \right)$$

$$= -\pi + \frac{\pi}{2} = -\frac{\pi}{2}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$= \frac{1}{\pi} \int_{-\pi}^0 -\pi \cos nx dx + \frac{1}{\pi} \int_0^{\pi} x \cos nx dx$$

$$= - \left[\frac{\sin nx}{n} \right]_{-\pi}^0 + \frac{1}{\pi} \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi}$$

$$= -(0 - 0) + \frac{1}{\pi} \left[0 + \frac{\cos n\pi}{n^2} - 0 - \frac{1}{n^2} \right]$$

$$= \frac{1}{\pi} \left[\frac{(-1)^n - 1}{n^2} \right] = \frac{1}{\pi n^2} [(-1)^n - 1]$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

$$= \frac{1}{\pi} \int_{-\pi}^0 -\pi \sin nx dx + \frac{1}{\pi} \int_0^{\pi} x \sin nx dx$$

$$b_n = -1 \left[\frac{-\cos nx}{n} \right]_{-\pi}^0 + \frac{1}{\pi} \left[\frac{-x \cos nx}{n} + \frac{\sin nx}{n^2} \right]_{\pi}^0$$

$$= \left(\frac{1}{n} - \frac{\cos n\pi}{n} \right) + \frac{1}{\pi} \left(-\frac{\pi \cos n\pi}{n} + 0 + 0 - 0 \right)$$

$$= \frac{1}{n} - \frac{\cos n\pi}{n} - \frac{\cos n\pi}{n} = \frac{1}{n} [1 - 2(-1)^n]$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

$$= \frac{-\pi/2}{2} + \sum_{n=1}^{\infty} \frac{1}{\pi n^2} [(-1)^n - 1] \cos nx + \frac{1}{n} [1 - 2(-1)^n] \sin nx$$

↳ ②

Take $x=0$

$$\textcircled{1} \Rightarrow [f(x)]_{x=0} = \frac{f_1(x=0) + f_2(x=0)}{2} = \frac{-\pi + 0}{2} = \frac{-\pi}{2} \rightarrow \textcircled{3}$$

$$\textcircled{2} \Rightarrow [f(x)]_{x=0} = \frac{-\pi}{4} + \sum_{n=1}^{\infty} \frac{1}{\pi n^2} [(-1)^n - 1] (1) + 0$$

$$= \frac{-\pi}{4} + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{[(-1)^n - 1]}{n^2} \rightarrow \textcircled{4}$$

From ③ & ④

$$\frac{-\pi}{2} = \frac{-\pi}{4} + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{[(-1)^n - 1]}{n^2}$$

$$\frac{1}{\pi} \sum_{n=1}^{\infty} \frac{[(-1)^n - 1]}{n^2} = \frac{-\pi}{2} + \frac{\pi}{4}$$

$$\frac{1}{\pi} \left[\frac{-2}{1^2} + 0 - \frac{2}{3^2} + 0 - \frac{2}{5^2} + \dots \right] = \frac{-\pi}{4}$$

$$\frac{-2}{\pi} \left[\frac{1}{1^2} + \frac{1}{3^2} + \dots \right] = \frac{-\pi}{4}$$

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$$

8) Find fourier series of $f(x) = \begin{cases} 1 + \frac{2x}{\pi}, & (-\pi, 0) \\ 1 - \frac{2x}{\pi}, & (0, \pi) \end{cases}$

Deduce $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \dots$

$$f(x) = \begin{cases} 1 + \frac{2x}{l}, & (-l, 0) \\ 1 - \frac{2x}{l}, & (0, l) \end{cases} \rightarrow \textcircled{1}$$

$$f_1(-x) = 1 + \frac{2(-x)}{l} = 1 - \frac{2x}{l} = f_2(x)$$

$\Rightarrow f(x)$ is even function

$$\therefore b_n = 0$$

$$a_0 = \frac{2}{l} \int_0^l f(x) dx$$

$$= \frac{2}{l} \int_0^l \left(1 - \frac{2x}{l}\right) dx$$

$$= \frac{2}{l} \left[x - \frac{2x^2}{2l} \right]_0^l$$

$$= \frac{2}{l} \left[l - \frac{l^2}{l} - 0 - 0 \right]$$

$$= 2 [1 - 1] = 0$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{l} \int_0^l \left(1 - \frac{2x}{l}\right) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{l^2} \int_0^l (l - 2x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{l^2} \left[\frac{(l - 2x) \sin\left(\frac{n\pi x}{l}\right)}{n\pi/l} + \frac{(-2) \cos\left(\frac{n\pi x}{l}\right)}{(n\pi/l)^2} \right]_0^l$$

$$= \frac{2}{l^2} \left[-\frac{l^2}{n\pi} \sin n\pi - \frac{2l^2}{(n\pi)^2} \cos n\pi - 0 + \frac{2l^2}{(n\pi)^2} (1) \right]$$

$$= \frac{2}{l^2} \left[-\frac{2l^2}{n^2\pi^2} (-1)^n + \frac{2l^2}{n^2\pi^2} (1) \right]$$

$$= \frac{4}{n^2\pi^2} [1 - (-1)^n]$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$f(x) = 0 + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} [1 - (-1)^n] \cos\left(\frac{n\pi x}{2}\right)$$

$$f(x) = \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} [1 - (-1)^n] \cos\left(\frac{n\pi x}{2}\right) \rightarrow (2)$$

Take $x=0$

$$(1) \Rightarrow [f(x)]_{x=0} = \frac{(1+0) + (1-0)}{2} = \frac{2}{2} = 1 \rightarrow (3)$$

$$(2) \Rightarrow [f(x)]_{x=0} = \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} [1 - (-1)^n] (1) \rightarrow (4)$$

From (3) & (4)

$$1 = \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} [1 - (-1)^n]$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} [1 - (-1)^n] = \frac{\pi^2}{4}$$

$$\frac{2}{1^2} + \frac{2}{3^2} + \frac{2}{5^2} + \dots = \frac{\pi^2}{4}$$

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$$

q) Find the fourier series expansion of

$$f(x) = \begin{cases} -x+1, & (-\pi, 0) \\ x+1, & (0, \pi) \end{cases} \rightarrow (1)$$

$$f_1(-x) = -(-x)+1 = x+1 = f_2(x)$$

It is an even function.

$$b_n = 0$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} (x+1) dx$$

$$= \frac{2}{\pi} \left[\frac{x^2}{2} + x \right]_0^{\pi} = \frac{2}{\pi} \left[\frac{\pi^2}{2} + \pi \right] = \frac{2}{\pi} \left(\frac{\pi^2 + 2\pi}{2} \right)$$

$$= 2 + \pi$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} (x+1) \cos nx dx$$

$$= \frac{2}{\pi} \left[\frac{(x+1) \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi}$$

$$a_n = \frac{2}{\pi} \left[0 + \frac{\cos n\pi}{n^2} - 0 - \frac{\cos(0)}{n^2} \right]$$

$$= \frac{2}{\pi} \left[\frac{(-1)^n}{n^2} - \frac{1}{n^2} \right]$$

$$= \frac{2}{\pi n^2} [(-1)^n - 1]$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

$$= 2 + \pi + \sum_{n=1}^{\infty} \frac{2}{\pi n^2} [(-1)^n - 1] \cos nx$$

$$f(x) = 2 + \pi + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{[(-1)^n - 1]}{n^2} \cos nx$$

10) Find FS expansion of $f(x)$ in $[-\pi, \pi]$,

$$\text{where } f(x) = \begin{cases} \pi + x, & -\pi \leq x \leq -\pi/2 \\ -x, & -\pi/2 \leq x \leq 0 \\ x, & 0 \leq x \leq \pi/2 \\ \pi - x, & \pi/2 \leq x \leq \pi. \end{cases}$$

$$f_1(x) = \pi + (-x) = \pi - x = f_4(x)$$

It is an even function. $\Rightarrow b_n = 0$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$= \frac{2}{\pi} \left[\int_0^{\pi/2} x dx + \int_{\pi/2}^{\pi} (\pi - x) dx \right]$$

$$= \frac{2}{\pi} \left(\frac{x^2}{2} \right)_0^{\pi/2} + \frac{2}{\pi} \left(\pi x - \frac{x^2}{2} \right)_{\pi/2}^{\pi}$$

$$= \frac{1}{\pi} \left(\frac{\pi^2}{4} - 0 \right) + \frac{2}{\pi} \left(\pi^2 - \frac{\pi^2}{2} - \frac{\pi^2}{2} + \frac{\pi^2}{8} \right)$$

$$= \frac{\pi}{4} + \frac{2}{\pi} \left(\frac{\pi^2}{8} \right) = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

$$= \frac{2}{\pi} \int_0^{\pi/2} x \cos nx dx + \frac{2}{\pi} \int_{\pi/2}^{\pi} (\pi - x) \cos nx dx$$

$$\begin{aligned}
 a_n &= \frac{2}{\pi} \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi/2} + \frac{2}{\pi} \left[\frac{(\pi-x) \sin nx}{n} - \frac{\cos nx}{n^2} \right]_{\pi/2}^{\pi} \\
 &= \frac{2}{\pi} \left[\frac{\pi}{2n} \sin\left(\frac{n\pi}{2}\right) + \frac{\cos\left(\frac{n\pi}{2}\right)}{n^2} - 0 - \frac{1}{n^2} \right] \\
 &\quad + \frac{2}{\pi} \left[0 - \frac{\cos n\pi}{n^2} - \frac{\pi}{2n} \sin\left(\frac{n\pi}{2}\right) + \frac{\cos\left(\frac{n\pi}{2}\right)}{n^2} \right] \\
 &= \frac{2}{\pi} \left[\frac{2}{n^2} \cos\left(\frac{n\pi}{2}\right) - \frac{(-1)^n}{n^2} - \frac{1}{n^2} \right] \\
 &= \frac{2}{\pi n^2} \left[2 \cos\left(\frac{n\pi}{2}\right) - (-1)^n - 1 \right]
 \end{aligned}$$

$$\begin{aligned}
 f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \\
 &= \frac{\pi/2}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2} \left[2 \cos\left(\frac{n\pi}{2}\right) - (-1)^n - 1 \right] \cos nx \\
 &= \frac{\pi}{4} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n^2} \left[2 \cos\left(\frac{n\pi}{2}\right) - (-1)^n - 1 \right] \cos nx.
 \end{aligned}$$

11) Find a fourier series for $f(x) = 1 - x^2$, $-1 \leq x \leq 1$.

$$f(x) = 1 - x^2 \rightarrow \textcircled{1}, \quad -1 \leq x \leq 1$$

$$f(-x) = 1 - (-x)^2 = 1 - x^2 = f(x)$$

It is an even function $\Rightarrow b_n = 0$, $l = 1$

$$a_0 = \frac{2}{l} \int_0^l f(x) dx = \frac{2}{1} \int_0^1 (1 - x^2) dx$$

$$= 2 \left[x - \frac{x^3}{3} \right]_0^1 = 2 \left[1 - \frac{1}{3} \right] = 2 \left(\frac{2}{3} \right) = \frac{4}{3}$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx = \frac{2}{1} \int_0^1 (1 - x^2) \cos n\pi x dx$$

$$= 2 \left[\frac{(1 - x^2) \sin n\pi x}{n\pi} - \frac{(-2x)(-\cos n\pi x)}{(n\pi)^2} + \frac{(-2)(-\sin n\pi x)}{(n\pi)^3} \right]_0^1$$

$$= 2 \left[0 - \frac{2 \cos n\pi}{(n\pi)^2} + 0 - 0 + 0 - 0 \right] = -\frac{4 \cos n\pi}{(n\pi)^2}$$

$$a_n = \frac{-4(-1)^n}{n^2 \pi^2}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$= \frac{4/3}{2} + \sum_{n=1}^{\infty} \frac{-4(-1)^n}{n^2 \pi^2} \cos(n\pi x)$$

$$f(x) = \frac{2}{3} - \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(n\pi x)$$

12) Find the fourier series for $f(x) = 1 + \sin x$,

$$-1 < x < 1$$

$$f(x) = 1 + \sin x$$

$f(x)$ is neither odd nor even.

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx = \frac{1}{1} \int_{-1}^1 (1 + \sin x) dx$$

$$= \left[x - \frac{\cos x}{1} \right]_{-1}^1 = [1 - \cos(1) + 1 + \cos(-1)]$$

$$= 2 - \cos(1) + \cos(1) = 2$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{1}{1} \int_{-1}^1 (1 + \sin x) \cos(n\pi x) dx$$

$$= \frac{(1 + \sin x) \sin(n\pi x)}{n}$$

$$= \int_{-1}^1 \cos n\pi x dx + \int_{-1}^1 \sin x \cos n\pi x dx$$

$$= \int_{-1}^1 \cos n\pi x dx + \frac{1}{2} \int_{-1}^1 [\sin((n\pi+1)x) - \sin((n\pi-1)x)] dx$$

$$= \left[\frac{\sin n\pi x}{n\pi} \right]_{-1}^1 + \frac{1}{2} \left[-\frac{\cos((n\pi+1)x)}{n\pi+1} + \frac{\cos((n\pi-1)x)}{n\pi-1} \right]_{-1}^1$$

$$= 0 + \frac{1}{2} \left[\frac{-\cos(n\pi+1)}{n\pi+1} + \frac{\cos(n\pi-1)}{n\pi-1} + \frac{\cos(n\pi+1)}{n\pi+1} - \frac{\cos(n\pi-1)}{n\pi-1} \right]$$

$$a_n = 0$$

$$\begin{aligned} b_n &= \frac{1}{2} \int_{-1}^1 f(x) \sin\left(\frac{n\pi x}{2}\right) dx = \frac{1}{2} \int_{-1}^1 (1 + \sin x) \sin n\pi x dx \\ &= \int_{-1}^1 \sin n\pi x dx + \int_{-1}^1 \sin x \sin n\pi x dx \\ &= \left[\frac{\cos n\pi x}{n\pi} \right]_{-1}^1 + \frac{1}{2} \int_{-1}^1 [\cos(n\pi-1)x - \cos(n\pi+1)x] dx \\ &= \left[\frac{\cos n\pi}{n\pi} - \frac{\cos n\pi}{n\pi} \right] + \frac{1}{2} \left[\frac{\sin(n\pi-1)x}{n\pi-1} - \frac{\sin(n\pi+1)x}{n\pi+1} \right]_{-1}^1 \\ &= 0 + \frac{1}{2} \left[\frac{\sin(n\pi-1)}{n\pi-1} - \frac{\sin(n\pi+1)}{n\pi+1} + \frac{\sin(n\pi-1)}{n\pi-1} - \frac{\sin(n\pi+1)}{n\pi+1} \right] \\ &= \frac{1}{2} \left[\frac{2\sin(n\pi-1)}{n\pi-1} - \frac{2\sin(n\pi+1)}{n\pi+1} \right] \\ &= \frac{\sin(n\pi-1)}{n\pi-1} - \frac{\sin(n\pi+1)}{n\pi+1} \end{aligned}$$

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{2}\right) + b_n \sin\left(\frac{n\pi x}{2}\right) \\ &= \frac{2}{2} + \sum_{n=1}^{\infty} 0 + \left[\frac{\sin(n\pi-1)}{n\pi-1} - \frac{\sin(n\pi+1)}{n\pi+1} \right] \sin(n\pi x) \\ f(x) &= 1 + \sum_{n=1}^{\infty} \left[\frac{\sin(n\pi-1)}{n\pi-1} - \frac{\sin(n\pi+1)}{n\pi+1} \right] \sin(n\pi x) \end{aligned}$$

13) Find f.s expansion for $f(x) = \begin{cases} 0, & -8 < x < 0 \\ 4, & 0 < x < 4 \\ 0, & 4 < x < 8 \end{cases}$
Neither odd nor even function.

$$\begin{aligned} a_0 &= \frac{1}{2} \int_{-8}^8 f(x) dx = \frac{1}{8} \int_{-8}^8 f(x) dx \\ &= \frac{1}{8} \int_0^4 4 dx = \frac{1}{2} (x)_0^4 = \frac{1}{2} (4-0) = 2 \end{aligned}$$

$$a_n = \frac{1}{2} \int_{-1}^1 f(x) \cos\left(\frac{n\pi x}{2}\right) dx$$

$$= \frac{1}{8} \int_0^4 4 \cos\left(\frac{n\pi x}{8}\right) dx$$

$$= \frac{1}{2} \left[\frac{\sin\left(\frac{n\pi x}{8}\right)}{n\pi/8} \right]_0^4$$

$$= \frac{8}{2n\pi} \left[\sin\left(\frac{4n\pi}{8}\right) - \sin(0) \right]$$

$$= \frac{4}{n\pi} \left[\sin\left(\frac{n\pi}{2}\right) - 0 \right]$$

$$= \frac{4}{n\pi} \sin\left(\frac{n\pi}{2}\right)$$

$$b_n = \frac{1}{2} \int_{-1}^1 f(x) \sin\left(\frac{n\pi x}{2}\right) dx$$

$$= \frac{1}{8} \int_0^4 4 \sin\left(\frac{n\pi x}{8}\right) dx$$

$$= \frac{1}{2} \left[-\frac{\cos\left(\frac{n\pi x}{8}\right)}{n\pi/8} \right]_0^4$$

$$= \frac{-8}{2n\pi} \left[\cos\left(\frac{n\pi}{2}\right) - 1 \right]$$

$$= \frac{4}{n\pi} \left[1 - \cos\left(\frac{n\pi}{2}\right) \right]$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{2}\right) + b_n \sin\left(\frac{n\pi x}{2}\right)$$

$$= \frac{2}{2} + \sum_{n=1}^{\infty} \frac{4}{n\pi} \sin\left(\frac{n\pi}{2}\right) \cos\left(\frac{n\pi x}{2}\right)$$

$$+ \frac{4}{n\pi} \left[1 - \cos\left(\frac{n\pi}{2}\right) \right] \sin\left(\frac{n\pi x}{2}\right)$$

$$f(x) = 1 + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi}{2}\right) \cos\left(\frac{n\pi x}{2}\right)$$

$$+ \frac{1}{n} \left[1 - \cos\left(\frac{n\pi}{2}\right) \right] \sin\left(\frac{n\pi x}{2}\right)$$

14) Obtain the FS for $f(x) = 1+x+x^2$, $(-\pi, \pi)$. D.T

$$\frac{1}{1^2} + \frac{1}{2^2} + \dots = \frac{\pi^2}{6}$$

$$f(x) = 1+x+x^2$$

Neither odd nor even

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (1+x+x^2) dx$$

$$= \frac{1}{\pi} \left[x + \frac{x^2}{2} + \frac{x^3}{3} \right]_{-\pi}^{\pi} = \frac{1}{\pi} \left[\pi + \frac{\pi^2}{2} + \frac{\pi^3}{3} + \pi - \frac{\pi^2}{2} + \frac{\pi^3}{3} \right]$$

$$= \frac{1}{\pi} \left[2\pi + \frac{2\pi^3}{3} \right] = 2 + \frac{2\pi^2}{3} = 2 \left(1 + \frac{\pi^2}{3} \right)$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (1+x+x^2) \cos nx dx$$

$$= \frac{1}{\pi} \left[\frac{(1+x+x^2) \sin nx}{n} - \frac{(1+2x)(-\cos nx)}{n^2} + \frac{(2)(-\sin nx)}{n^3} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[0 + \frac{(1+2\pi) \cos n\pi}{n^2} - 0 - 0 - \frac{(1+2\pi) \cos n\pi}{n^2} + 0 \right]$$

$$= \frac{1}{\pi} \left[\frac{\cos n\pi}{n^2} + \frac{2\pi \cos n\pi}{n^2} + \frac{\cos n\pi}{n^2} + \frac{2\pi \cos n\pi}{n^2} \right]$$

$$= \frac{(-1)^n}{\pi} \left[\frac{2}{n^2} + \frac{4\pi}{n^2} \right] = \frac{(-1)^n}{\pi n^2} (2+4\pi)$$

$$= \frac{2(-1)^n}{\pi n^2} (1+2\pi)$$

$$a_n = 0$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (1+x+x^2) \sin nx dx$$

$$= \frac{1}{\pi} \left[\frac{-(1+x+x^2) \cos nx}{n} - \frac{(1+2x)(-\sin nx)}{n^2} + \frac{(2)(\cos nx)}{n^3} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{-(1+\pi+\pi^2) \cos n\pi}{n} - 0 + \frac{2 \cos n\pi}{n^3} + \frac{(1-\pi+\pi^2) \cos n\pi}{n} - 0 - \frac{2 \cos n\pi}{n^3} \right]$$

$$= \frac{1}{\pi} \left[\frac{-\cos n\pi}{n} - \frac{\pi \cos n\pi}{n} - \frac{\pi^2 \cos n\pi}{n} + \frac{\cos n\pi}{n} - \frac{\pi \cos n\pi}{n} + \frac{\pi^2 \cos n\pi}{n} \right]$$

$$= \frac{1}{\pi} \left[\frac{-2\pi \cos n\pi}{n} \right]$$

$$= \frac{-2(-1)^n}{n}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

$$= \frac{2(1 + \frac{\pi^3}{3})}{2} + \sum_{n=1}^{\infty} 0 - \frac{2(-1)^n}{n} \sin nx$$

$$f(x) = 1 + \frac{\pi^3}{3} - 2 \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin nx$$

15) If 'a' is neither zero nor an integer, find the F.S expansion of period 2π for $f(x) = \sin ax$, in $-\pi \leq x \leq \pi$.

$$f(x) = \sin ax$$

$$f(-x) = \sin(-ax) = -\sin ax = -f(x)$$

$\therefore f(x)$ is an odd function.

$$\Rightarrow a_0 = a_n = 0$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \sin ax \sin nx dx$$

$$= \frac{2}{\pi} \left(\frac{1}{2} \right) \int_0^{\pi} [\cos(n-a)x - \cos(n+a)x] dx$$

$$= \frac{1}{\pi} \left[\frac{\sin(n-a)x}{n-a} - \frac{\sin(n+a)x}{n+a} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{\sin(n\pi - a\pi)}{n-a} - \frac{\sin(n\pi + a\pi)}{n+a} \right]$$

$$= \frac{1}{\pi} \left[\frac{-(-1)^n \sin a\pi}{a-n-a} - \frac{(-1)^n \sin a\pi}{n+a} \right]$$

$$= \frac{-(-1)^n \sin a\pi}{\pi} \left[\frac{1}{n-a} + \frac{1}{n+a} \right]$$

$$= \frac{-(-1)^n \sin a\pi}{\pi} \left[\frac{n+a+n-a}{n^2-a^2} \right]$$

$$= \frac{(-1)^n \sin a\pi}{\pi (a^2-n^2)} (2n) = \frac{2n(-1)^n \sin a\pi}{\pi (a^2-n^2)}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

$$= \sum_{n=1}^{\infty} \frac{2n(-1)^n \sin a\pi}{\pi (a^2-n^2)} \sin nx$$

$$f(x) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n \sin a\pi}{(a^2-n^2)} \sin nx.$$

1b) Obtain F.S for $f(x) = |\cos x|$ in $(-\pi, \pi)$.

$$f(x) = |\cos x| \rightarrow \textcircled{1} \quad |\cos x| = \begin{cases} \cos x, & (0, \pi) \\ -\cos x, & (-\pi, 0) \end{cases}$$

It is an even function.

$$\therefore b_n = 0$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi/2} \cos x dx + \frac{2}{\pi} \int_{\pi/2}^{\pi} (-\cos x) dx$$

$$= \frac{2}{\pi} \left[\sin x \right]_0^{\pi/2} - \frac{2}{\pi} \left[\sin x \right]_{\pi/2}^{\pi}$$

$$= \frac{2}{\pi} [1-0] - \frac{2}{\pi} [0-1] = \frac{2}{\pi} + \frac{2}{\pi} = \frac{4}{\pi}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

$$= \frac{2}{\pi} \int_0^{\pi/2} \cos x \cos nx dx + \frac{2}{\pi} \int_{\pi/2}^{\pi} (-\cos x) \cos nx dx$$

$$= \frac{2}{\pi} \left(\frac{1}{2} \right) \int_0^{\pi/2} [\cos(n+1)x + \cos(n-1)x] dx - \frac{2}{\pi} \left(\frac{1}{2} \right) \int_{\pi/2}^{\pi} [\cos(n+1)x + \cos(n-1)x] dx$$

$$\begin{aligned}
 a_n &= \frac{1}{\pi} \left[\frac{\sin(n+1)x}{n+1} + \frac{\sin(n-1)x}{n-1} \right]_{\pi/2} \\
 &\quad - \frac{1}{\pi} \left[\frac{\sin(n+1)x}{n+1} + \frac{\sin(n-1)x}{n-1} \right]_{\pi/2}^{\pi} \\
 &= \frac{1}{\pi} \left[\frac{\sin(\frac{n\pi}{2} + \frac{\pi}{2})}{n+1} + \frac{\sin(\frac{n\pi}{2} - \frac{\pi}{2})}{n-1} \right] \\
 &\quad - \frac{1}{\pi} \left[(0+0) - \frac{\sin(\frac{n\pi}{2} + \frac{\pi}{2})}{n+1} - \frac{\sin(\frac{n\pi}{2} - \frac{\pi}{2})}{n-1} \right] \\
 &= \frac{1}{\pi} \left[\frac{2 \sin(\frac{n\pi}{2} + \frac{\pi}{2})}{n+1} + \frac{2 \sin(\frac{n\pi}{2} - \frac{\pi}{2})}{n-1} \right] \\
 &= \frac{2}{\pi} \left[\frac{\sin(\frac{\pi}{2} + \frac{n\pi}{2})}{n+1} - \frac{\sin(\frac{\pi}{2} - \frac{n\pi}{2})}{n-1} \right] \\
 &= \frac{2}{\pi} \left[\frac{\cos(\frac{n\pi}{2})}{n+1} - \frac{\cos(\frac{n\pi}{2})}{n-1} \right] \\
 &= \frac{2}{\pi} \cos\left(\frac{n\pi}{2}\right) \left[\frac{n-1-n-1}{n^2-1} \right] \\
 &= -\frac{4}{\pi} \cos\left(\frac{n\pi}{2}\right) \left(\frac{1}{n^2-1}\right) \\
 &= -\frac{4}{\pi(n^2-1)} \cos\left(\frac{n\pi}{2}\right)
 \end{aligned}$$

$$\begin{aligned}
 a_1 &= \frac{2}{\pi} \int_0^{\pi} f(x) \cos x \, dx = \frac{2}{\pi} \int_0^{\pi/2} (\cos x) \cos x \, dx \\
 &\quad - \frac{2}{\pi} \int_{\pi/2}^{\pi} \cos x \cos x \, dx \\
 &= \frac{2}{\pi} \int_0^{\pi/2} \cos^2 x \, dx - \frac{2}{\pi} \int_{\pi/2}^{\pi} \cos^2 x \, dx \\
 &= \frac{2}{\pi} \int_0^{\pi/2} \left(\frac{1+\cos 2x}{2} \right) dx - \frac{2}{\pi} \int_{\pi/2}^{\pi} \left(\frac{1+\cos 2x}{2} \right) dx \\
 &= \frac{1}{\pi} \left(x + \frac{\sin 2x}{2} \right)_{\pi/2}^{\pi} - \frac{1}{\pi} \left(x + \frac{\sin 2x}{2} \right)_{\pi/2}^{\pi}
 \end{aligned}$$

$$= \frac{1}{\pi} \left[\frac{\pi}{2} + 0 - 0 - 0 \right] - \frac{1}{\pi} \left[\cancel{2}\pi + 0 - \frac{\pi}{2} - 0 \right]$$

$$= \frac{1}{2} - \frac{1}{2} = 0$$

$$b_1 = \frac{2}{\pi} \int_0^{\pi} f(x) \sin x \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi/2} \cos x \sin x \, dx - \frac{2}{\pi} \int_{\pi/2}^{\pi} \cos x \sin x \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi/2} \frac{\sin 2x}{2} \, dx - \frac{2}{\pi} \int_{\pi/2}^{\pi} \frac{\sin 2x}{2} \, dx$$

$$= \frac{1}{\pi} \left[\frac{-\cos 2x}{2} \right]_0^{\pi/2} - \frac{1}{\pi} \left[\frac{-\cos 2x}{2} \right]_{\pi/2}^{\pi}$$

$$= -\frac{1}{2\pi} (-1 - 1) + \frac{1}{2\pi} (1 - (-1))$$

$$= \frac{2}{2\pi} + \frac{2}{2\pi} = \frac{4}{2\pi} = \frac{2}{\pi}$$

$$f(x) = \frac{a_0}{2} + a_1 \cos x + b_1 \sin x + \sum_{n=2}^{\infty} a_n \cos nx + b_n \sin nx$$

$$= \frac{4/\pi}{2} + 0 + \frac{2}{\pi} \sin x + \sum_{n=2}^{\infty} \frac{-4}{\pi(n^2-1)} \cos\left(\frac{n\pi}{2}\right) \cos nx$$

$$f(x) = \frac{2}{\pi} + \frac{2 \sin x}{\pi} - \frac{4}{\pi} \sum_{n=2}^{\infty} \frac{1}{(n^2-1)} \cos\left(\frac{n\pi}{2}\right) \cos nx$$

Problems on full range

$(0, 2\pi)$ & $(0, 2l)$

17) Find F.S for $f(x) = \begin{cases} x, & 0 \leq x \leq 3 \\ 6-x, & 3 \leq x \leq 6 \end{cases}$

Sol:

$$a_0 = \frac{1}{l} \int_0^{2l} f(x) dx$$

Here $2l = 6 \Rightarrow l = 3$

$$\begin{aligned} a_0 &= \frac{1}{3} \int_0^6 f(x) dx = \frac{1}{3} \int_0^3 x dx + \frac{1}{3} \int_3^6 (6-x) dx \\ &= \frac{1}{3} \left[\frac{x^2}{2} \right]_0^3 + \frac{1}{3} \left[6x - \frac{x^2}{2} \right]_3^6 \\ &= \frac{1}{3} \left(\frac{9}{2} \right) + \frac{1}{3} \left(36 - 18 - 18 + \frac{9}{2} \right) \\ &= \frac{3}{2} + \frac{1}{3} \left(\frac{9}{2} \right) = \frac{3}{2} + \frac{3}{2} = 3 \end{aligned}$$

$$\begin{aligned} a_n &= \frac{1}{l} \int_0^{2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx \\ &= \frac{1}{3} \left[\int_0^3 x \cos\left(\frac{n\pi x}{3}\right) dx + \int_3^6 (6-x) \cos\left(\frac{n\pi x}{3}\right) dx \right] \\ &= \frac{1}{3} \left\{ \left[\frac{x \sin\left(\frac{n\pi x}{3}\right)}{n\pi/3} + \frac{\cos\left(\frac{n\pi x}{3}\right)}{(n\pi/3)^2} \right]_0^3 \right. \\ &\quad \left. + \left[\frac{(6-x) \sin\left(\frac{n\pi x}{3}\right)}{n\pi/3} - \frac{\cos\left(\frac{n\pi x}{3}\right)}{(n\pi/3)^2} \right]_3^6 \right\} \\ &= \frac{1}{3} \left\{ \left[\frac{3 \sin n\pi}{n\pi/3} + \frac{\cos n\pi}{(n\pi/3)^2} - 0 - \frac{9}{n^2 \pi^2} \right] \right. \\ &\quad \left. + \left[0 - \frac{\cos 2n\pi}{(n\pi/3)^2} - 0 + \frac{\cos n\pi}{(n\pi/3)^2} \right] \right\} \\ &= \frac{1}{3} \left[\frac{9(-1)^n}{n^2 \pi^2} - \frac{9}{n^2 \pi^2} - \frac{9}{n^2 \pi^2} + \frac{9(-1)^n}{(n\pi/3)^2} \right] \end{aligned}$$

$$= 3 \left[\frac{2(-1)^n}{n^2 \pi^2} - \frac{2}{n^2 \pi^2} \right]$$

$$= \frac{6}{n^2 \pi^2} [(-1)^n - 1]$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{1}{3} \left\{ \int_0^3 x \sin\left(\frac{n\pi x}{3}\right) dx + \int_3^6 (6-x) \sin\left(\frac{n\pi x}{3}\right) dx \right\}$$

$$= \frac{1}{3} \left\{ \left[\frac{-x \cos\left(\frac{n\pi x}{3}\right)}{n\pi/3} + \frac{\sin\left(\frac{n\pi x}{3}\right)}{(n\pi/3)^2} \right]_0^3 + \left[\frac{-(6-x) \cos\left(\frac{n\pi x}{3}\right)}{n\pi/3} - \frac{\sin\left(\frac{n\pi x}{3}\right)}{(n\pi/3)^2} \right]_3^6 \right\}$$

$$= \frac{1}{3} \left\{ \left[-\frac{3 \cos n\pi}{n\pi/3} + 0 + 0 - 0 \right] + \left[0 - 0 + \frac{3 \cos n\pi}{n\pi/3} - 0 \right] \right\}$$

$$b_n = 0$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l}$$

$$= \frac{3}{2} + \sum_{n=1}^{\infty} \frac{6}{n^2 \pi^2} [(-1)^n - 1] \cos\left(\frac{n\pi x}{3}\right) + 0$$

$$f(x) = \frac{3}{2} + \frac{6}{\pi^2} \sum_{n=1}^{\infty} \frac{[(-1)^n - 1]}{n^2} \cos\left(\frac{n\pi x}{3}\right)$$

18) Find f.s of $f(x) = 2x - x^2$ in $0 < x < 2$.

Sol: Here $2l = 2 \Rightarrow l = 1$

$$a_0 = \frac{1}{l} \int_0^{2l} f(x) dx = \frac{1}{1} \int_0^2 (2x - x^2) dx$$

$$= \left[\frac{2x^2}{2} - \frac{x^3}{3} \right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$$

$$a_n = \frac{1}{l} \int_0^{2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{1}{1} \int_0^2 (2x - x^2) \cos n\pi x dx$$

$$= \left[\frac{(2x - x^2) \sin n\pi x}{n\pi} - \frac{(2 - 2x)(\cos n\pi x)}{(n\pi)^2} + \frac{(-2)(-\sin n\pi x)}{(n\pi)^3} \right]_0^2$$

$$= \left[0 - \frac{(-2)(-\cos 2n\pi)}{(n\pi)^2} + 0 \right.$$

$$\left. - 0 + \frac{(2)(-1)}{(n\pi)^2} - 0 \right]$$

$$= \frac{-2}{(n\pi)^2} - \frac{2}{(n\pi)^2} = \frac{-4}{n^2 \pi^2}$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{1}{1} \int_0^2 (2x - x^2) \sin n\pi x dx$$

$$= \left[-\frac{(2x - x^2) \cos n\pi x}{n\pi} - \frac{(2 - 2x)(-\sin n\pi x)}{(n\pi)^2} + \frac{(-2)(\cos n\pi x)}{(n\pi)^3} \right]_0^2$$

$$= \left(0 - 0 - \frac{2 \cos 2n\pi}{(n\pi)^3} \right) - \left(0 - 0 - \frac{2}{(n\pi)^3} \right)$$

$$= \frac{-2}{(n\pi)^3} + \frac{2}{(n\pi)^3} = 0$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$f(x) = \frac{4/3}{2} + \sum_{n=1}^{\infty} \frac{-4}{(n\pi)^2} \cos n\pi x$$

$$f(x) = \frac{2}{3} - \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \cos n\pi x$$

19) Obtain F.S of periodicity 2π for $f(x) = e^{-x}$ in the interval $0 < x < 2\pi$. Hence deduce

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{1+n^2} = \frac{\pi}{e^{\pi} - e^{-\pi}} \quad \text{Further derive a series}$$

for cosech π

Ans a sol:

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx = \frac{1}{\pi} \int_0^{2\pi} e^{-x} dx$$

$$= \frac{1}{\pi} [-e^{-x}]_0^{2\pi} = \frac{1}{\pi} [1 - e^{-2\pi}]$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{2\pi} e^{-x} \cos nx dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-x}}{1+n^2} (-\cos nx + n \sin nx) \right]_0^{2\pi}$$

$$= \frac{1}{\pi(1+n^2)} [e^{-2\pi}(-1) - 1(-1)]$$

$$= \frac{1 - e^{-2\pi}}{\pi(1+n^2)}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} e^{-x} \sin nx dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-x}}{1+n^2} (-\sin nx - n \cos nx) \right]_0^{2\pi}$$

$$= \frac{1}{\pi(1+n^2)} [e^{-2\pi}(-n) - 1(-n)]$$

$$= \frac{n[1 - e^{-2\pi}]}{\pi(1+n^2)}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

$$f(x) = \left(\frac{1 - e^{-2\pi}}{\pi} \right) + \sum_{n=1}^{\infty} \frac{1 - e^{-2\pi}}{\pi(1+n^2)} \cos nx + \frac{n[1 - e^{-2\pi}]}{\pi(1+n^2)} \sin nx$$

$$e^{-\pi} = \frac{1 - e^{-2\pi}}{\pi} \left\{ \frac{1}{2} + \sum_{n=1}^{\infty} \frac{\cos nx}{(1+n^2)} + \frac{n \sin nx}{(1+n^2)} \right\}$$

Put $x = \pi$

$$e^{-\pi} = \frac{1 - e^{-2\pi}}{\pi} \left\{ \frac{1}{2} + \sum_{n=1}^{\infty} \frac{(-1)^n}{(1+n^2)} \right\}$$

$$\frac{1}{2} - \frac{1}{2} + \sum_{n=2}^{\infty} \frac{(-1)^n}{1+n^2} = \frac{\pi e^{-\pi}}{1 - e^{-2\pi}}$$

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{(1+n^2)} = \frac{\pi}{e^{\pi} - e^{-\pi}} \rightarrow (i)$$

$$\operatorname{cosech} \pi = \frac{2}{e^{\pi} - e^{-\pi}}$$

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{(1+n^2)} = \frac{1}{2} \left[\frac{2\pi}{e^{\pi} - e^{-\pi}} \right]$$

$$\frac{1}{2} \pi \operatorname{cosech} \pi = \sum_{n=2}^{\infty} \frac{(-1)^n}{(1+n^2)}$$

$$\operatorname{cosech} \pi = \frac{2}{\pi} \sum_{n=2}^{\infty} \frac{(-1)^n}{(1+n^2)}$$

20) Expand in fourier series of $f(x) = x \sin x$
 $0 < x < 2\pi$. Deduce the result $\hookrightarrow (1)$

$$\frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots = \frac{\pi - 2}{4}$$

$$\text{Sol: } a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx = \frac{1}{\pi} \int_0^{2\pi} x \sin x dx$$

$$a_0 = \frac{1}{\pi} \left[\frac{-x \cos x}{1} + \frac{\sin x}{1} \right]_0^{2\pi}$$

$$a_0 = \frac{1}{\pi} [-2\pi \cos 2\pi + 0 + 0 - 0] = -2$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx \, dx = \frac{1}{\pi} \int_0^{2\pi} x \sin x \cos nx \, dx$$

$$= \frac{1}{2\pi} \int_0^{2\pi} x [\sin(n+1)x - \sin(n-1)x] \, dx$$

$$= \frac{1}{2\pi} \left[\frac{-x \cos(n+1)x}{n+1} + \frac{\sin(n+1)x}{(n+1)^2} \right]_0^{2\pi}$$

$$- \frac{1}{2\pi} \left[\frac{-x \cos(n-1)x}{n-1} + \frac{\sin(n-1)x}{(n-1)^2} \right]_0^{2\pi}$$

$$= \frac{1}{2\pi} \left[\frac{-2\pi \cos(2\pi + 2n\pi)}{n+1} + \frac{\sin(2\pi + 2n\pi)}{(n+1)^2} + 0 - 0 \right]$$

$$- \frac{1}{2\pi} \left[\frac{-2\pi \cos(2\pi - 2n\pi)}{n-1} + \frac{\sin(2\pi - 2n\pi)}{(n-1)^2} + 0 - 0 \right]$$

$$= \frac{1}{2\pi} \left[\frac{-2\pi}{n+1} + \frac{2\pi}{n-1} \right] = + \left[\frac{-1}{n+1} + \frac{1}{n-1} \right] = - \left[\frac{-n+1+n+1}{n^2-1} \right]$$

$$a_n = \frac{+2}{n^2-1}$$

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx \\
 &= \frac{1}{\pi} \int_0^{2\pi} x \sin x \sin nx \, dx \\
 &= \frac{1}{2\pi} \int_0^{2\pi} x [\cos(n-1)x - \cos(n+1)x] \, dx \\
 &= \frac{1}{2\pi} \left[\frac{x \sin(n-1)x}{n-1} + \frac{\cos(n-1)x}{(n-1)^2} \right]_0^{2\pi} \\
 &\quad - \frac{1}{2\pi} \left[\frac{x \sin(n+1)x}{n+1} + \frac{\cos(n+1)x}{(n+1)^2} \right]_0^{2\pi} \\
 &= \frac{1}{2\pi} \left[\frac{2\pi \sin(2n\pi - 2\pi)}{n-1} + \frac{\cos(2\pi - 2n\pi)}{(n-1)^2} - 0 - \frac{1}{(n-1)^2} \right] \\
 &\quad - \frac{1}{2\pi} \left[0 + \frac{1}{(n+1)^2} - \frac{2\pi \sin(2\pi + 2n\pi)}{n+1} \right. \\
 &\quad \left. + \frac{\cos(2\pi + 2n\pi)}{(n+1)^2} - 0 - \frac{1}{(n+1)^2} \right] \\
 &= \frac{1}{2\pi} \left[\frac{1}{(n-1)^2} - \frac{1}{(n-1)^2} + \frac{1}{(n+1)^2} - \frac{1}{(n+1)^2} \right] \\
 &= 0
 \end{aligned}$$

$$a_1 = \frac{1}{\pi} \int_0^{2\pi} x \sin x \cos x \, dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} x \frac{\sin 2x}{2} \, dx$$

$$= \frac{1}{2\pi} \left[-\frac{x \cos 2x}{2} + \frac{\sin 2x}{4} \right]_0^{2\pi}$$

$$= \frac{1}{2\pi} \left[-\frac{2\pi \cos 4\pi}{2} + \frac{\sin 4\pi}{4} - 0 - 0 \right]$$

$$= -\frac{1}{2}$$

$$b_1 = \frac{1}{\pi} \int_0^{2\pi} x \sin x \sin x \, dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} x \sin^2 x \, dx = \frac{1}{\pi} \int_0^{2\pi} x \frac{(1 - \cos 2x)}{2} \, dx$$

$$\begin{aligned}
 b_1 &= \frac{1}{2\pi} \int_0^{2\pi} [x - x \cos 2x] dx \\
 &= \frac{1}{2\pi} \left(\frac{x^2}{2} \right)_0^{2\pi} - \frac{1}{2\pi} \left[\frac{x \sin 2x}{2} + \frac{\cos 2x}{4} \right]_0^{2\pi} \\
 &= \frac{1}{2\pi} \left(\frac{4\pi^2}{2} \right) - \frac{1}{2\pi} \left[0 + \frac{1}{4} - 0 - \frac{1}{4} \right] \\
 &= \pi
 \end{aligned}$$

$$\begin{aligned}
 f(x) &= \frac{a_0}{2} + a_1 \cos x + b_1 \sin x + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \\
 &= \frac{-2}{2} + \left(-\frac{1}{2} \right) \cos x + \pi \sin x + \sum_{n=2}^{\infty} \frac{+2}{n^2-1} \cos nx \\
 &= -1 - \frac{1}{2} \cos x + \pi \sin x + \sum_{n=2}^{\infty} \frac{2}{n^2-1} \cos nx \rightarrow (2)
 \end{aligned}$$

Take $x = \frac{\pi}{2}$

$$(1) \Rightarrow [f(x)]_{x=\frac{\pi}{2}} = \frac{\pi}{2} \sin\left(\frac{\pi}{2}\right) = \frac{\pi}{2} \rightarrow (3)$$

$$(2) \Rightarrow [f(x)]_{x=\frac{\pi}{2}} = -1 - 0 + \pi + \sum_{n=2}^{\infty} \frac{2}{n^2-1} \cos\left(\frac{n\pi}{2}\right) \rightarrow (4)$$

(3) & (4)

$$\frac{\pi}{2} = -1 + \pi + 2 \sum_{n=2}^{\infty} \frac{1}{(n-1)(n+1)} \cos\left(\frac{n\pi}{2}\right)$$

$$2 \sum_{n=2}^{\infty} \frac{1}{(n-1)(n+1)} \cos\left(\frac{n\pi}{2}\right) = \frac{\pi}{2} + 1 - \pi$$

$$2 \left[\frac{(-1)}{1 \cdot 3} + 0 + \frac{(1)}{3 \cdot 5} + 0 + \frac{(-1)}{5 \cdot 7} + \dots \right] = 1 - \frac{\pi}{2}$$

$$-2 \left[\frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots \right] = 1 - \frac{\pi}{2}$$

$$\frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots = \frac{-1}{2} + \frac{\pi}{4}$$

$$\frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots = \frac{\pi - 2}{4}$$

21) If $f(x) = \begin{cases} \sin x, & 0 < x < \pi \\ 0, & \pi < x < 2\pi \end{cases}$. Find a F.S. $\rightarrow \textcircled{1}$

of periodicity 2π and evaluate

$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots$$

Sol:

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} \sin x dx$$

$$= \frac{1}{\pi} (-\cos x)_0^{\pi} = \frac{1}{\pi} (1+1) = \frac{2}{\pi}$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx$$

$$= \frac{1}{\pi} \int_0^{\pi} \sin x \cos nx dx$$

$$= \frac{1}{2\pi} \left[\int_0^{\pi} \sin(x+nx) dx + \int_0^{\pi} \sin(x-nx) dx \right]$$

$$= \frac{1}{2\pi} \int_0^{\pi} \sin(n+1)x dx - \frac{1}{2\pi} \int_0^{\pi} \sin(n-1)x dx$$

$$= \frac{1}{2\pi} \left[\frac{\cos(n+1)x}{n+1} \right]_0^{\pi} + \frac{1}{2\pi} \left[\frac{\cos(n-1)x}{n-1} \right]_0^{\pi}$$

$$= \frac{-1}{2\pi(n+1)} [\cos(n+1)\pi - 1]$$

$$+ \frac{1}{2\pi(n-1)} [\cos(n-1)\pi - 1]$$

$$= \frac{-1}{2\pi(n+1)} [-(-1)^n - 1] + \frac{1}{2\pi(n-1)} [-(-1)^n - 1]$$

$$= \frac{1}{2\pi(n+1)} [+(-1)^n + 1] - \frac{1}{2\pi(n-1)} [(-1)^n + 1]$$

$$= \frac{1+(-1)^n}{2\pi} \left[\frac{1}{n+1} - \frac{1}{n-1} \right]$$

$$= \frac{[1+(-1)^n]}{2\pi} \left[\frac{n-1-n-1}{n^2-1} \right] = \frac{-2[1+(-1)^n]}{\pi(n^2-1)}$$

$$a_n = \frac{-[(-1)^n + 1]}{\pi(n^2-1)}$$

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx = \frac{1}{\pi} \int_0^{\pi} \sin x \sin nx \, dx \\
 &= \frac{1}{2\pi} \int_0^{\pi} \cos(n-1)x - \cos(n+1)x \, dx \\
 &= \frac{1}{2\pi} \left[\frac{\sin(n-1)x}{n-1} - \frac{\sin(n+1)x}{n+1} \right]_0^{\pi} \\
 &= \frac{1}{2\pi} [0 - 0 - 0 + 0] = 0
 \end{aligned}$$

$$\begin{aligned}
 a_1 &= \frac{1}{\pi} \int_0^{2\pi} f(x) \cos x \, dx = \frac{1}{\pi} \int_0^{\pi} \sin x \cos x \, dx \\
 &= \frac{1}{2\pi} \int_0^{\pi} \sin 2x \, dx = \frac{1}{2\pi} \left[-\frac{\cos 2x}{2} \right]_0^{\pi} \\
 &= -\frac{1}{4\pi} [1 - 1] = 0
 \end{aligned}$$

$$\begin{aligned}
 b_1 &= \frac{1}{\pi} \int_0^{2\pi} f(x) \sin x \, dx = \frac{1}{\pi} \int_0^{\pi} \sin x \sin x \, dx \\
 &= \frac{1}{\pi} \int_0^{\pi} \sin^2 x \, dx = \frac{1}{2\pi} \int_0^{\pi} (1 - \cos 2x) \, dx \\
 &= \frac{1}{2\pi} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi} = \frac{1}{2\pi} [\pi - 0 - 0 - 0] = \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 f(x) &= \frac{a_0}{2} + a_1 \cos x + a_2 \sin x + \sum_{n=2}^{\infty} a_n \cos nx + b_n \sin nx \\
 &= \frac{2/\pi}{2} + 0 + \frac{1}{2} \sin x + \sum_{n=2}^{\infty} \frac{-[(-1)^n + 1]}{\pi(n^2 - 1)} \cos nx \rightarrow 0 \\
 &= \frac{1}{\pi} + \frac{1}{2} \sin x - \frac{1}{\pi} \sum_{n=2}^{\infty} \frac{1 + (-1)^n}{n^2 - 1} \cos nx \rightarrow (2)
 \end{aligned}$$

Take $x=0$

$$(1) \Rightarrow [f(x)]_{x=0} = 0 \rightarrow (3)$$

$$(2) \Rightarrow [f(x)]_{x=0} = \frac{1}{\pi} - \frac{1}{\pi} \sum_{n=2}^{\infty} \frac{[1 + (-1)^n]}{n^2 - 1} (1) \rightarrow (4)$$

From (3) & (4)

$$0 = \frac{1}{\pi} - \frac{1}{\pi} \sum_{n=2}^{\infty} \frac{[1 + (-1)^n]}{n^2 - 1}$$

$$\frac{1}{\pi} \sum_{n=2}^{\infty} \frac{[1+(-1)^n]}{(n-1)(n+1)} = \frac{1}{\pi}$$

$$\frac{2}{1.3} + 0 + \frac{2}{3.5} + 0 + \frac{2}{5.7} + \dots = 1$$

$$2 \left[\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots \right] = 1$$

$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots = \frac{1}{2}$$

22) An alternating current after passing through a rectifier has the form $i = \begin{cases} i_0 \sin x, & 0 \leq x \leq \pi, \\ 0, & \pi \leq x \leq 2\pi \end{cases}$ where i_0 is maximum current and the period is 2π

Sol $i = \begin{cases} i_0 \sin x, & 0 \leq x \leq \pi \\ 0, & \pi \leq x \leq 2\pi \end{cases} \rightarrow \textcircled{1}$

$$A_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx = \frac{1}{\pi} \int_0^{2\pi} i dx$$

$$= \frac{1}{\pi} \int_0^{\pi} i_0 \sin x dx$$

$$= \frac{i_0}{\pi} [-\cos x]_0^{\pi} = -\frac{i_0}{\pi} [-1 - 1]$$

$$= \frac{2i_0}{\pi}$$

$$A_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx$$

$$= \frac{1}{\pi} \int_0^{\pi} i_0 \sin x \cos nx dx$$

$$= \frac{i_0}{\pi} \int_0^{\pi} \frac{1}{2} [\sin(n+1)x - \sin(n-1)x] dx$$

$$= \frac{i_0}{2\pi} \left[-\frac{\cos(n+1)x}{n+1} + \frac{\cos(n-1)x}{n-1} \right]_0^{\pi}$$

$$= \frac{i_0}{2\pi} \left[\frac{-\cos(n\pi+\pi)}{n+1} + \frac{\cos(n\pi-\pi)}{n-1} + \frac{1}{n+1} - \frac{1}{n-1} \right]$$

$$= \frac{i_0}{2\pi} \left[\frac{(-1)^n}{n+1} - \frac{(-1)^n}{n-1} + \frac{1}{n+1} - \frac{1}{n-1} \right]$$

$$a_n = \frac{i_0}{2\pi} \left[(-1)^n \left[\frac{1}{n+1} - \frac{1}{n-1} \right] + \frac{1}{n+1} - \frac{1}{n-1} \right]$$

$$= \frac{i_0}{2\pi} \left[(-1)^n \left(\frac{n-1-n-1}{n^2-1} \right) + \frac{n-1-n-1}{n^2-1} \right]$$

$$= \frac{i_0}{2\pi} \left[\frac{-2(-1)^n}{n^2-1} + \frac{-2}{n^2-1} \right]$$

$$= \frac{-i_0}{2\pi(n^2-1)} [1 + (-1)^n]$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} i \sin nx \, dx = \frac{1}{\pi} \int_0^{\pi} i_0 \sin x \sin nx \, dx$$

$$= \frac{i_0}{2\pi} \int_0^{\pi} \sin(n+1)x + \sin \cos(n-1)x + \cos(n+1)x \, dx$$

$$= \frac{i_0}{2\pi} \left[\frac{\sin(n-1)x}{n-1} - \frac{\sin(n+1)x}{n+1} \right]_0^{\pi}$$

$$= \frac{i_0}{2\pi} [0 - 0 - 0 + 0] = 0$$

$$a_1 = \frac{1}{\pi} \int_0^{2\pi} i \cos x \, dx = \frac{1}{\pi} \int_0^{\pi} i_0 \sin x \cos x \, dx$$

$$= \frac{i_0}{2\pi} \int_0^{\pi} \sin 2x \, dx = \frac{i_0}{2\pi} \left[-\frac{\cos 2x}{2} \right]_0^{\pi}$$

$$= \frac{-i_0}{4\pi} [1 - 1] = 0$$

$$b_1 = \frac{1}{\pi} \int_0^{2\pi} i \sin x \, dx = \frac{1}{\pi} \int_0^{\pi} i_0 \sin x \sin x \, dx$$

$$= \frac{i_0}{\pi} \int_0^{\pi} \sin^2 x \, dx = \frac{i_0}{2\pi} \int_0^{\pi} (1 - \cos 2x) \, dx$$

$$= \frac{i_0}{2\pi} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi} = \frac{i_0}{2\pi} [\pi - 0 - 0 + 0]$$

$$= \frac{i_0}{2}$$

$$f(x) = \frac{a_0}{2} + a_1 \cos x + b_1 \sin x + \sum_{n=2}^{\infty} a_n \cos nx + b_n \sin nx$$

$$= \frac{2i_0}{2\pi} + 0 + \frac{i_0}{2} \sin x + \sum_{n=2}^{\infty} \frac{-i_0 [1 + (-1)^n]}{\pi(n^2-1)} \cos nx$$

$$f(x) = \frac{l_0}{\pi} + \frac{l_0}{2} \sin x - \frac{l_0}{\pi} \sum_{n=2}^{\infty} \frac{[1+(-1)^n]}{n^2-1} \cos nx$$

23) Find the F.S of period 2π for the function

$$f(x) = \frac{1}{12} x(\pi-x)(2\pi-x) \text{ in } (0, 2\pi). \text{ Deduce}$$

the sum of series $\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \dots$

$$f(x) = \frac{1}{12} x(\pi-x)(2\pi-x)$$

$$= \frac{1}{12} (\pi x - x^2)(2\pi - x)$$

$$= \frac{1}{12} (2\pi^2 x - \pi x^2 - 2\pi x^2 + x^3)$$

$$f(x) = \frac{1}{12} (x^3 - 3\pi x^2 + 2\pi^2 x) \rightarrow \textcircled{1}$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} \frac{1}{12} (x^3 - 3\pi x^2 + 2\pi^2 x) dx$$

$$= \frac{1}{12\pi} \left[\frac{x^4}{4} - \frac{3\pi x^3}{3} + \frac{2\pi^2 x^2}{2} \right]_0^{2\pi}$$

$$= \frac{1}{12\pi} \left[\frac{16\pi^4}{4} - \pi(8\pi^3) + \pi^2(4\pi^2) - 0 \right]$$

$$= \frac{1}{12\pi} [4\pi^4 - 8\pi^4 + 4\pi^4]$$

$$a_0 = 0$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx$$

$$= \frac{1}{12\pi} \int_0^{2\pi} (x^3 - 3\pi x^2 + 2\pi^2 x) \cos nx dx$$

$$= \frac{1}{12\pi} \left[\int_0^{2\pi} x^3 \cos nx dx \right.$$

$$\left. - 3\pi \int_0^{2\pi} x^2 \cos nx dx \right.$$

$$\left. + 2\pi^2 \int_0^{2\pi} x \cos nx dx \right]$$

$$= \frac{1}{12\pi} [I_1 - 3\pi I_2 + 2\pi^2 I_3]$$

$$I_1 = \int_0^{2\pi} x^3 \cos nx \, dx$$

$$= \left[\frac{x^3 \sin nx}{n} - \frac{3x^2 (-\cos nx)}{n^2} + \frac{6x (-\sin nx)}{n^3} \right]_0^{2\pi}$$

$$= 0 + \frac{12\pi^2}{n^2} \cos 2n\pi + 0 - 0 = 0$$

$$I_1 = \int_0^{2\pi} x^3 \cos nx \, dx$$

$$= \left[\frac{x^3 \sin nx}{n} - \frac{3x^2 (-\cos nx)}{n^2} + \frac{6x (-\sin nx)}{n^3} - \frac{6 \cos nx}{n^4} \right]_0^{2\pi}$$

$$= 0 + \frac{12\pi^2}{n^2} (1) + 0 - \frac{6}{n^4} - 0 - 0 - 0 + \frac{6}{n^4}$$

$$= \frac{12\pi^2}{n^2}$$

$$I_2 = \int_0^{2\pi} x^2 \cos nx \, dx$$

$$= \left[\frac{x^2 \sin nx}{n} - \frac{2x (\cos nx)}{n^2} + \frac{2 (-\sin nx)}{n^3} \right]_0^{2\pi}$$

$$= 0 + \frac{4\pi \cos(2n\pi)}{n^2} - 0 - 0 - 0 + 0$$

$$= \frac{4\pi}{n^2}$$

$$I_3 = \int_0^{2\pi} x \cos nx \, dx = \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{2\pi}$$

$$= 0 + \frac{1}{n^2} - 0 - \frac{1}{n^2} = 0$$

$$a_n = \frac{1}{12\pi} \left[\frac{12\pi^2}{n^2} - 3\pi \left(\frac{4\pi}{n^2} \right) + 0 \right]$$

$$= \frac{1}{12\pi} \left[\frac{12\pi^2}{n^2} - \frac{12\pi^2}{n^2} \right] = 0$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx$$

$$= \frac{1}{12\pi} \int_0^{2\pi} (x^3 - 3\pi x^2 + 2\pi^2 x) \sin nx \, dx$$

$$= \frac{1}{12\pi} \left[\int_0^{2\pi} x^3 \sin nx \, dx - 3\pi \int_0^{2\pi} x^2 \sin nx \, dx + 2\pi^2 \int_0^{2\pi} x \sin nx \, dx \right]$$

$$= \frac{1}{12\pi} [I_1 - 3\pi I_2 + 2\pi^2 I_3]$$

$$I_1 = \int_0^{2\pi} x^3 \sin nx \, dx$$

$$= \left[\frac{-x^3 \cos nx}{n} - \frac{3x^2(-\sin nx)}{n^2} + \frac{6x(\cos nx)}{n^3} - \frac{6 \sin nx}{n^4} \right]_0^{2\pi}$$

$$= \frac{-8\pi^3}{n} + 0 + \frac{12\pi}{n^3} - 0 - 0 - 0 - 0 - 0$$

$$= \frac{12\pi}{n^3} - \frac{8\pi^3}{n}$$

$$I_2 = \int_0^{2\pi} x^2 \sin nx \, dx$$

$$= \left[\frac{-x^2 \cos nx}{n} - \frac{2x(-\sin nx)}{n^2} + \frac{2 \cos nx}{n^3} \right]_0^{2\pi}$$

$$= \frac{-4\pi^2}{n} + 0 + \frac{2}{n^3} - 0 - 0 - \frac{2}{n^3}$$

$$= -\frac{4\pi^2}{n}$$

$$I_3 = \int_0^{2\pi} x \sin nx \, dx$$

$$= \left[\frac{-x \cos nx}{n} + \frac{\sin nx}{n} \right]_0^{2\pi}$$

$$= \frac{-2\pi}{n} + 0 - 0 - 0 = -\frac{2\pi}{n}$$

$$\begin{aligned}
 b_n &= \frac{1}{12\pi} \left[\frac{12\pi}{n^3} - \frac{8\pi^3}{n} - 3\pi \left(-\frac{4\pi^2}{n} \right) + 2\pi^2 \left(-\frac{2\pi}{n} \right) \right] \\
 &= \frac{1}{12\pi} \left[\frac{12\pi}{n^3} - \frac{8\pi^3}{n} + \frac{12\pi^3}{n} - \frac{4\pi^3}{n} \right] \\
 &= \frac{1}{12\pi} \left(\frac{12\pi}{n^3} + \frac{12\pi^3}{n} - \frac{4\pi^3}{n} \right) = \frac{1}{n^3}
 \end{aligned}$$

$$\begin{aligned}
 f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \\
 &= 0 + \sum_{n=1}^{\infty} 0 + \frac{1}{n^3} \sin nx = \sum_{n=1}^{\infty} \frac{\sin nx}{n^3} \rightarrow (2)
 \end{aligned}$$

$$x = \frac{\pi}{2}$$

$$\begin{aligned}
 (1) \Rightarrow [f(x)]_{x=\frac{\pi}{2}} &= \frac{1}{12} \frac{\pi}{2} \left(\pi - \frac{\pi}{2} \right) \left(2\pi - \frac{\pi}{2} \right) \\
 &= \frac{1}{12} \frac{\pi}{2} \left(\frac{\pi}{2} \right) \left(\frac{3\pi}{2} \right) = \frac{3\pi^3}{12 \times 8} = \frac{\pi^3}{32} \rightarrow (3)
 \end{aligned}$$

$$(2) \Rightarrow [f(x)]_{x=\frac{\pi}{2}} = \sum_{n=1}^{\infty} \frac{1}{n^3} \sin \left(\frac{n\pi}{2} \right) \rightarrow (4)$$

From (3) & (4)

$$\frac{\pi^3}{32} = \sum_{n=1}^{\infty} \frac{1}{n^3} \sin \left(\frac{n\pi}{2} \right)$$

$$\frac{1}{1^3} (1) + \frac{1}{3^3} (-1) + \frac{1}{5^3} (1) + \dots = \frac{\pi^3}{32}$$

$$\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \dots = \frac{\pi^3}{32}$$

24) Derive the Fourier expansion of $\sqrt{1-\cos x}$ in $0 \leq x \leq 2\pi$ & Deduce that $\sum \frac{1}{4n^2-1} = \frac{1}{2}$.

$$f(x) = \sqrt{1-\cos x}$$

$$\therefore f(x) = \sqrt{2} \sin \left(\frac{x}{2} \right), \quad 0 \leq x \leq 2\pi$$

$\hookrightarrow (1)$

$$\sqrt{1-\cos x}$$

$$\sin^2 A = \frac{1-\cos 2A}{2}$$

$$2 \sin^2 \frac{A}{2} = 1 - \cos A$$

$$\Rightarrow \sqrt{2 \sin^2 \left(\frac{x}{2} \right)} = \pm \sqrt{2} \sin \left(\frac{x}{2} \right)$$

$$\begin{aligned}
 a_0 &= \frac{1}{\pi} \int_0^{2\pi} f(x) dx = \frac{1}{\pi} \int_0^{2\pi} \sqrt{2} \sin\left(\frac{x}{2}\right) dx \\
 &= \frac{\sqrt{2}}{\pi} \left[\frac{-\cos(x/2)}{1/2} \right]_0^{2\pi} \\
 &= -\frac{\sqrt{2}}{\pi} \left[\frac{\cos \pi}{1/2} - \frac{\cos(0)}{1/2} \right] \\
 &= -\frac{2\sqrt{2}}{\pi} (-1-1) = \frac{4\sqrt{2}}{\pi}
 \end{aligned}$$

$$\begin{aligned}
 a_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \\
 &= \frac{1}{\pi} \int_0^{2\pi} \sqrt{2} \sin\left(\frac{x}{2}\right) \cos nx dx \\
 &= \frac{\sqrt{2}}{2\pi} \int_0^{2\pi} \sin\left(n+\frac{1}{2}\right)x - \sin\left(n-\frac{1}{2}\right)x dx \\
 &= \frac{\sqrt{2}}{2\pi} \left[\frac{-\cos\left(n+\frac{1}{2}\right)x}{n+\frac{1}{2}} + \frac{\cos\left(n-\frac{1}{2}\right)x}{n-\frac{1}{2}} \right]_0^{2\pi} \\
 &= \frac{\sqrt{2}}{2\pi} \left[\frac{-\cos(2n\pi+\pi)}{n+\frac{1}{2}} + \frac{\cos(2n\pi-\pi)}{n-\frac{1}{2}} \right. \\
 &\quad \left. + \frac{1}{n+\frac{1}{2}} - \frac{1}{n-\frac{1}{2}} \right] \\
 &= \frac{\sqrt{2}}{2\pi} \left[\frac{1}{n+\frac{1}{2}} - \frac{1}{n-\frac{1}{2}} + \frac{1}{n+\frac{1}{2}} - \frac{1}{n-\frac{1}{2}} \right] \\
 &= \frac{\sqrt{2}}{\pi} \left[\frac{1}{n+\frac{1}{2}} - \frac{1}{n-\frac{1}{2}} \right] \\
 &= \frac{\sqrt{2}}{\pi} \left[\frac{n-\frac{1}{2} - n - \frac{1}{2}}{(n^2 - \frac{1}{4})} \right] \\
 &= \frac{\sqrt{2}}{\pi} \left[\frac{(-1)}{n^2 - \frac{1}{4}} \right] \\
 &= \frac{-4\sqrt{2}}{\pi(4n^2-1)}
 \end{aligned}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} \sqrt{2} \sin\left(\frac{x}{2}\right) \sin nx \, dx$$

$$= \frac{\sqrt{2}}{2\pi} \int_0^{2\pi} \cos\left(n - \frac{1}{2}\right)x - \cos\left(n + \frac{1}{2}\right)x \, dx$$

$$= \frac{\sqrt{2}}{2\pi} \left[\frac{\sin\left(n - \frac{1}{2}\right)x}{n - \frac{1}{2}} - \frac{\sin\left(n + \frac{1}{2}\right)x}{n + \frac{1}{2}} \right]_0^{2\pi}$$

$$= \frac{\sqrt{2}}{2\pi} \left[\frac{\sin(2n\pi - \pi)}{n - \frac{1}{2}} - \frac{\sin(2n\pi + \pi)}{n + \frac{1}{2}} - 0 + 0 \right]$$

$$= \frac{\sqrt{2}}{2\pi} [0 - 0] = 0$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

$$= \frac{4\sqrt{2}/\pi}{2} + \sum_{n=1}^{\infty} \frac{-4\sqrt{2}}{\pi(4n^2-1)} \cos nx$$

$$f(x) = \frac{2\sqrt{2}}{\pi} - \frac{4\sqrt{2}}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2-1} \cos nx \rightarrow (2)$$

Take $x=0$

$$(1) \Rightarrow [f(x)]_{x=0} = \frac{f(0) + f(2\pi)}{2} = 0 \rightarrow (3)$$

$$(2) \Rightarrow [f(x)]_{x=0} = \frac{2\sqrt{2}}{\pi} - \frac{4\sqrt{2}}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2-1} \quad (1) \rightarrow (4)$$

(3) & (4),

$$0 = \frac{2\sqrt{2}}{\pi} - \frac{4\sqrt{2}}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2-1} \cos nx$$

$$\frac{4\sqrt{2}}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2-1} \cos nx = \frac{2\sqrt{2}}{\pi}$$

$$\sum_{n=1}^{\infty} \frac{1}{4n^2-1} = \frac{1}{2}$$

Parseval's Relation

25) Prove that in the interval $0 < x < l$,

$$x = \frac{l}{2} - \frac{4l}{\pi^2} \left[\cos \frac{\pi x}{l} + \frac{1}{3^2} \cos \frac{3\pi x}{l} + \dots \right]$$

Deduce $\frac{1}{1^4} + \frac{1}{3^4} + \dots = \frac{\pi^4}{96}$.

Given : $f(x) = x$, $(0, l)$

Half range cosine series.

$$b_n = 0$$

$$\begin{aligned} a_0 &= \frac{2}{l} \int_0^l f(x) dx = \frac{2}{l} \int_0^l x dx \\ &= \frac{2}{l} \left(\frac{x^2}{2} \right)_0^l = \frac{1}{l} (l^2) = l. \end{aligned}$$

$$\begin{aligned} a_n &= \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx \\ &= \frac{2}{l} \int_0^l x \cos\left(\frac{n\pi x}{l}\right) dx \\ &= \frac{2}{l} \left[\frac{x \sin(n\pi x/l)}{n\pi/l} + \frac{\cos(n\pi x/l)}{(n\pi/l)^2} \right]_0^l \\ &= \frac{2}{l} \left[0 + \frac{l^2}{n^2 \pi^2} \cos n\pi - 0 - \frac{l^2}{n^2 \pi^2} \right] \\ &= \frac{2l}{n^2 \pi^2} [(1)^n - 1] \end{aligned}$$

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \\ &= \frac{l}{2} + \sum_{n=1}^{\infty} \frac{2l}{n^2 \pi^2} [(1)^n - 1] \cos\left(\frac{n\pi x}{l}\right) \\ &= \frac{l}{2} + \frac{2l}{\pi^2} \left[\frac{-2}{1^2} \cos\left(\frac{\pi x}{l}\right) + 0 \right. \\ &\quad \left. - \frac{2}{3^2} \cos\left(\frac{3\pi x}{l}\right) + \dots \right] \end{aligned}$$

$$x = \frac{l}{2} - \frac{4l}{\pi^2} \left[\cos\left(\frac{\pi x}{l}\right) + \frac{1}{3^2} \cos\left(\frac{3\pi x}{l}\right) + \dots \right]$$

Parseval's identity:

$$\frac{1}{U.L-L.L} \int_{U.L-L.L}^{\overset{U.L}{f(x)}} [f(x)]^2 dx = \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} a_n^2 + b_n^2$$

$$LHS \Rightarrow \frac{1}{l-0} \int_0^l x^2 dx = \frac{1}{l} \left(\frac{x^3}{3} \right)_0^l = \frac{1}{l} \left(\frac{l^3}{3} \right) = \frac{l^2}{3}$$

$$RHS \Rightarrow \frac{l^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{2l}{n^2 \pi^2} \right)^2 ([(-1)^n - 1])^2 + 0$$

$$\Rightarrow \frac{l^2}{4} + \frac{1}{2} \left(\frac{4l^2}{\pi^4} \right) \sum_{n=1}^{\infty} \frac{1}{n^4} [1 + 1 - 2(-1)^n]$$

$$\Rightarrow \frac{l^2}{4} + \frac{1}{2} \left(\frac{4l^2}{\pi^4} \right) \sum_{n=1}^{\infty} \frac{1}{n^4} 2 [1 - (-1)^n]$$

$$\Rightarrow \frac{l^2}{4} + \frac{4l^2}{\pi^4} \sum_{n=1}^{\infty} \frac{[1 - (-1)^n]}{n^4}$$

$$LHS = RHS$$

$$\Rightarrow \frac{l^2}{3} = \frac{l^2}{4} + \frac{4l^2}{\pi^4} \sum_{n=1}^{\infty} \frac{[1 - (-1)^n]}{n^4}$$

$$\frac{4l^2}{\pi^4} \sum_{n=1}^{\infty} \frac{[1 - (-1)^n]}{n^4} = \frac{l^2}{3} - \frac{l^2}{4}$$

$$\frac{4l^2}{\pi^4} \left[\frac{2}{1^4} + 0 + \frac{2}{3^4} + 0 + \frac{2}{5^4} + \dots \right] = \frac{l^2}{12}$$

$$\frac{8}{\pi^4} \left[\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots \right] = \frac{1}{12}$$

$$\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots = \frac{\pi^4}{96}$$

2b) obtain the half range cosine series for $f(x) = x$ in $(0, \pi)$ and hence prove that $\frac{1}{1^4} + \frac{1}{3^4} + \dots = \frac{\pi^4}{96}$.

Half range cosine series $\Rightarrow b_n = 0$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} x dx = \frac{2}{\pi} \left(\frac{x^2}{2} \right)_0^{\pi}$$

$$a_0 = \frac{2}{\pi} \left[\frac{\pi^2}{2} \right] = \pi$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \cos nx \, dx$$

$$= \frac{2}{\pi} \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[0 + \frac{\cos n\pi}{n^2} - 0 - \frac{1}{n^2} \right]$$

$$= \frac{2}{\pi} \left[\frac{(-1)^n}{n^2} - \frac{1}{n^2} \right] = \frac{2}{\pi n^2} [(-1)^n - 1]$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

$$= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2} [(-1)^n - 1] \cos nx$$

$$= \frac{\pi}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n^2} [(-1)^n - 1] \cos nx$$

Convergence theorem failed

Parseval's identity:

$$\frac{1}{L} \int_{-L}^L [f(x)]^2 \, dx = \frac{a_0^2}{4} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

$$\frac{1}{\pi - 0} \int_0^{\pi} x^2 \, dx = \frac{\pi^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{2}{n\pi} \right)^2 [(-1)^n - 1]^2$$

$$\frac{1}{\pi} \left(\frac{x^3}{3} \right)_0^{\pi} = \frac{\pi^2}{4} + \frac{4}{2\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^4} [2 - 2(-1)^n]$$

$$\frac{1}{\pi} \left(\frac{\pi^3}{3} \right) = \frac{\pi^2}{4} + \frac{2}{\pi^2} \sum_{n=1}^{\infty} \frac{2}{n^4} [1 - (-1)^n]$$

$$\frac{\pi^2}{3} = \frac{\pi^2}{4} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{[1 - (-1)^n]}{n^4}$$

$$\frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{[1 - (-1)^n]}{n^4} = \frac{\pi^2}{3} - \frac{\pi^2}{4}$$

$$\frac{4}{\pi^2} \left[\frac{2}{1^4} + 0 + \frac{2}{3^4} + 0 + \frac{2}{5^4} + \dots \right] = \frac{\pi^2}{12}$$

$$\frac{8}{\pi^2} \left[\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots \right] = \frac{\pi^2}{12}$$

$$\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots = \frac{\pi^4}{96}$$

Q7) obtain the sine series for $f(x) = c$ in $(0, \pi)$. Deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{8}$.

Sol: $f(x) = c \rightarrow$ ①

Half range Sine Series $\Rightarrow a_0 = a_n = 0$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} c dx = \frac{2c}{\pi} (x)_0^{\pi} = \frac{2c}{\pi} (\pi - 0)$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx = \frac{2}{\pi} \int_0^{\pi} c \sin nx dx$$

$$= \frac{2c}{\pi} \left[-\frac{\cos nx}{n} \right]_0^{\pi} = -\frac{2c}{n\pi} [\cos n\pi - \cos 0]$$

$$= -\frac{2c}{n\pi} [(1)^n - 1] = \frac{2c}{n\pi} [1 - (-1)^n]$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

$$= \sum_{n=1}^{\infty} \frac{2c}{n\pi} [1 - (-1)^n] \sin nx$$

$$f(x) = \frac{2c}{\pi} \sum_{n=1}^{\infty} \frac{[1 - (-1)^n]}{n} \sin nx \rightarrow$$
 ②

Convergence theorem fails. So we have to use Parseval's theorem.

$$\frac{1}{\text{U.L-L.L}} \int_{\text{L.L}}^{\text{U.L}} [f(x)]^2 dx = \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} a_n^2 + b_n^2$$

$$\frac{1}{\pi - 0} \int_0^{\pi} (c)^2 dx = \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{2c}{n\pi} \right)^2 [1 - (-1)^n]^2$$

$$\frac{1}{\pi} c^2 (x)_0^{\pi} = \frac{1}{2} \left(\frac{4c^2}{\pi^2} \right) \sum_{n=1}^{\infty} \frac{1}{n^2} [1 - (-1)^n]^2$$

$$\frac{C^2}{\pi} (\pi - 0) = \frac{4C^2}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} [1 - (-1)^n]$$

$$\frac{\pi^2}{4} = \sum_{n=1}^{\infty} \frac{1}{n^2} [1 - (-1)^n]$$

$$\frac{1}{1^2} (2) + 0 + \frac{1}{3^2} (2) + \dots = \frac{\pi^2}{4}$$

$$\frac{1}{1^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{8}$$

28) Find the fourier series for $f(x) = x^2$, $(-l, l)$ and Deduce that $\frac{1}{1^4} + \frac{1}{2^4} + \dots = \frac{\pi^4}{90}$.

$$f(x) = x^2 \rightarrow \textcircled{1}$$

Even function $\Rightarrow b_n = 0$

$$a_0 = \frac{2}{l} \int_0^l f(x) dx = \frac{2}{l} \int_0^l x^2 dx$$

$$= \frac{2}{l} \left(\frac{x^3}{3} \right)_0^l = \frac{2}{l} \left(\frac{l^3}{3} \right) = \frac{2l^2}{3}$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{l} \int_0^l x^2 \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{l} \left[\frac{x^2 \sin(n\pi x/l)}{n\pi/l} - \frac{2x [-\cos(n\pi x/l)]}{(n\pi/l)^2} \right]$$

$$+ \frac{2 [-\sin(n\pi x/l)]}{(n\pi/l)^3} \Bigg]_0^l$$

$$= \frac{2}{l} \left[0 + \frac{2l(l^2)}{(n\pi)^2} \cos n\pi - 0 - 0 - 0 \right]$$

$$= \frac{4l^2}{n^2 \pi^2} (-1)^n$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

$$= \frac{2l^2/3}{2} + \sum_{n=1}^{\infty} \frac{4l^2(-1)^n}{n^2 \pi^2} \cos nx$$

$$f(x) = \frac{l^2}{3} + \frac{4l^2}{\pi^2} \sum_{n=1}^{\infty} (-1)^n \cos nx \rightarrow (2)$$

Convergence theorem failed. So it is now we have to use Parseval's relation.

$$\frac{1}{L-L} \int_{L-L}^{L-L} [f(x)]^2 dx = \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} a_n^2 + b_n^2$$

$$\frac{1}{l+l} \int_{-l}^l (x^2)^2 dx = \left(\frac{2l^2}{3} \right)^2 + \frac{1}{2} \sum_{n=1}^{\infty} \left[\frac{4l^2}{n^2 \pi^2} \right]^2 [(-1)^n]^2$$

$$\frac{1}{2l} \int_{-l}^l x^4 dx = \frac{4l^4}{9 \times 4} + \frac{1}{2} \left(\frac{16l^4}{\pi^4} \right) \sum_{n=1}^{\infty} \frac{1}{n^4}$$

$$\frac{1}{2l} \left[\frac{x^5}{5} \right]_{-l}^l = \frac{l^4}{9} + \frac{8l^4}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{n^4}$$

$$\frac{l^4}{9} + \frac{8l^4}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{1}{2l} \left(\frac{l^5}{5} + \frac{l^5}{5} \right)$$

$$\frac{l^4}{9} + \frac{8l^4}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{1}{2l} \left(\frac{2l^5}{5} \right)$$

$$\frac{8l^4}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{l^4}{5} - \frac{l^4}{9}$$

$$\frac{8l^4}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{4l^4}{45}$$

$$\frac{2}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{1}{45}$$

$$\frac{1}{1^4} + \frac{1}{2^4} + \dots = \frac{\pi^4}{90}$$

Complex form of Fourier series.

29) Obtain the complex form of F.S for $f(x) = e^x$
 $(-\pi < x < \pi)$

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{inx}$$

$$C_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^x e^{-inx} dx$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{(1-in)x} dx = \frac{1}{2\pi} \left[\frac{e^{(1-in)x}}{1-in} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{2\pi(1-in)} \left[e^{(1-in)\pi} - e^{(1-in)(-\pi)} \right]$$

$$= \frac{1}{2\pi(1-in)} \left[e^{\pi} e^{-in\pi} - e^{-\pi} e^{in\pi} \right]$$

$$= \frac{1}{2\pi(1-in)} \left[-e^{-in\pi} + e^{in\pi} \right]$$

$$= \frac{2 \sinh a\pi}{2\pi(1-in)} = \frac{\sinh a\pi}{\pi(1-in)} \times \frac{(1+in)}{(1+in)}$$

$$= \frac{(1+in) \sinh a\pi}{\pi(1+n^2)}$$

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{(1+in) \sinh a\pi}{\pi(1+n^2)} e^{inx}$$

30) Find the complex form of F.S of periodicity π where $f(x) = \sin(kx)$ in $(-\pi < x < \pi)$

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{inx}$$

$$C_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-inx} \sin(kx) dx$$

$$= \frac{1}{2\pi} \left[\frac{e^{-inx}}{-n^2+k^2} \left[-in \sin(kx) - k \cos(kx) \right] \right]_{-\pi}^{\pi}$$

$$\begin{aligned}
 C_n &= \frac{1}{2\pi(k^2 - n^2)} \left[e^{-in\pi} (-in \sin k\pi - k \cos k\pi) \right. \\
 &\quad \left. - e^{in\pi} (in \sin k\pi - k \cos k\pi) \right] \\
 &= \frac{1}{2\pi(k^2 - n^2)} \left[(-1)^n \{-in \sin k\pi - k \cos k\pi - in \sin k\pi \right. \\
 &\quad \left. + k \cos k\pi\} \right] \\
 &= \frac{(-1)^n (-2in \sin k\pi)}{2\pi(k^2 - n^2)} = -\frac{(-1)^n in \sin k\pi}{\pi(k^2 - n^2)} \\
 &= \frac{i(-1)^n n \sin k\pi}{\pi(n^2 - k^2)}
 \end{aligned}$$

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{i(-1)^n n \sin k\pi}{\pi(n^2 - k^2)} e^{inx}$$

31) If a is not an integer, obtain the complex form of F.S in $f(x) = \cos ax$, $(-\pi < x < \pi)$

Sol: $f(x) = \sum_{n=-\infty}^{\infty} C_n e^{inx}$

$$C_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-inx} \cos ax dx$$

$$= \frac{1}{2\pi} \left[\frac{e^{-inx}}{a^2 - n^2} (-in \cos ax + a \sin ax) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{2\pi} \times \frac{1}{a^2 - n^2} \left[e^{-in\pi} (-in \cos a\pi + a \sin a\pi) \right. \\
 \left. - e^{in\pi} (-in \cos a\pi - a \sin a\pi) \right]$$

$$C_n = \frac{(-1)^n}{2\pi} \times \frac{1}{a^2 - n^2} \left[-in \cos a\pi + a \sin a\pi + in \cos a\pi + a \sin a\pi \right]$$

$$= \frac{(-1)^n 2a \sin a\pi}{2\pi(a^2 - n^2)}$$

$$C_n = \frac{a(-1)^n \sin a\pi}{\pi(a^2 - n^2)}$$

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{a(-1)^n \sin a\pi}{\pi(a^2 - n^2)} e^{inx}$$

32) Obtain the complex form of the F.S for $f(x) = \sin 2x$ in $-\pi < x < \pi$.

$$f(x) = \sin 2x \rightarrow \textcircled{1}$$

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{inx}$$

$$C_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-inx} \sin 2x dx$$

$$= \frac{1}{2\pi} \left[\frac{e^{-inx}}{-n^2 + 4} (-in \sin 2x - 2 \cos 2x) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{2\pi} \times \frac{1}{4 - n^2} \left[e^{-in\pi} (-in \sin 2\pi - 2 \cos 2\pi) - e^{in\pi} (in \sin 2\pi - 2 \cos 2\pi) \right]$$

$$= \frac{1}{2\pi} \times \frac{1}{(4 - n^2)} \left[(-1)^n (0 - 2 \cos 2\pi) - (-1)^n (0 - 2 \cos 2\pi) \right]$$

$$= \frac{1}{2\pi} \times \frac{1}{4 - n^2} \left[(-1)^n (-2 \cos 2\pi) - (-1)^n (-2 \cos 2\pi) \right]$$

$$C_n = \frac{-in(-1)^n \sin 2\pi}{\pi(4 - n^2)} = \frac{in(-1)^n \sin 2\pi}{\pi(n^2 - 4)}$$

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{in(-1)^n \sin 2\pi}{\pi(n^2-4)} e^{inx}$$

$$f(x) = \frac{i \sin 2\pi}{\pi} \sum_{n=-\infty}^{\infty} \frac{n(-1)^n}{(n^2-4)} e^{inx}$$

Harmonic Analysis:

33) Determine the first 2 harmonic of the F.S for the following values.

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$
y	1.98	1.30	1.05	1.30	-0.88	0.25

Here the 1st and last values are not repeated.
 $n=6$.

y	x	$y \cos x$	$y \sin x$	$y \cos 2x$	$y \sin 2x$
1.98	0	1.98	0	1.98	0
1.30	$\frac{\pi}{3} = 60^\circ$	0.65	1.13	-0.65	1.13
1.05	$\frac{2\pi}{3} = 120^\circ$	-0.53	0.91	-0.53	-0.91
1.30	$\pi = 180^\circ$	-1.3	0	1.3	0
-0.88	$\frac{4\pi}{3} = 240^\circ$	0.44	0.76	0.44	-0.76
0.25	$\frac{5\pi}{3} = 300^\circ$	0.13	-0.22	-0.13	-0.22
Σy = 5	—	$\Sigma y \cos x$ = 1.37	$\Sigma y \sin x$ = 2.58	$\Sigma y \cos 2x$ = 2.41	$\Sigma y \sin 2x$ = -0.76

$$a_0 = \frac{2}{n} \times \Sigma y = \frac{2}{6} \times 5 = 1.67$$

$$a_1 = \frac{2}{n} \times \Sigma y \cos x = \frac{2}{6} \times 1.37 = 0.46$$

$$a_2 = \frac{2}{n} \times \Sigma y \cos 2x = \frac{2}{6} \times 2.41 = 0.80$$

$$b_1 = \frac{2}{n} \times \Sigma y \sin x = \frac{2}{6} \times 2.58 = 0.86$$

$$b_2 = \frac{2}{n} \times \Sigma y \sin 2x = \frac{2}{6} \times (-0.76) = -0.25$$

$$f(x) = \frac{a_0}{2} + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x$$

$$= \frac{1.67}{2} + 0.46 \cos x + 0.86 \sin x + 0.8 \cos 2x - 0.25 \sin 2x$$

$$f(x) = 0.84 + 0.46 \cos x + 0.86 \sin x + 0.8 \cos 2x - 0.25 \sin 2x.$$

34) Obtain the constant term and first 2 harmonic of F.S for $f(x)$ in the following table.

x	0	1	2	3	4	5	6	7	8	9	10	11	12
$f(x)$	18	18.7	17.6	15	11.6	8.3	6	5.3	6.4	9	12.4	15	18

Here 1st and last values are not repeated.
So we have to include a column with $x=12$ & $f(x)=18$.

Here $2l=12 \Rightarrow l=6$ and $n=12$.

$f(x)$ (y)	x	$y \cos(\frac{\pi x}{6})$	$y \sin(\frac{\pi x}{6})$	$y \cos(\frac{2\pi x}{6})$	$y \sin(\frac{2\pi x}{6})$
18	0	18	0	18	0
18.7	1	16.19	9.35	9.35	16.19
17.6	2	8.8	15.24	-8.8	15.24
15	3	0	15	-15	0
11.6	4	-5.8	10.05	-5.8	-10.05
8.3	5	-7.19	4.15	4.15	-7.19
6	6	-6	0	6	0
5.3	7	-4.59	-2.65	2.65	4.59
6.4	8	-3.2	-5.54	-3.2	5.54
9	9	0	-9	-9	0
12.4	10	6.2	-10.74	6.2	-10.74
15	11	12.99	-7.5	7.5	-12.99
$\Sigma y =$ 143.3	-	$\Sigma y \cos(\frac{\pi x}{6})$ = 35.4	$\Sigma y \sin(\frac{\pi x}{6})$ = 18.36	$\Sigma y \cos(\frac{2\pi x}{6})$ = -0.35	$\Sigma y \sin(\frac{2\pi x}{6})$ = 0.59

$$f(x) = \frac{a_0}{2} + a_1 \cos\left(\frac{\pi x}{6}\right) + b_1 \sin\left(\frac{\pi x}{6}\right) + a_2 \cos\left(\frac{2\pi x}{6}\right) + b_2 \sin\left(\frac{2\pi x}{6}\right)$$

$$a_0 = \frac{2}{n} \times \sum y = \frac{2}{12} \times 143.3 = 23.88$$

$$a_1 = \frac{2}{n} \times \sum y \cos\left(\frac{\pi x}{6}\right) = \frac{2}{12} \times 35.4 = 5.9$$

$$b_1 = \frac{2}{n} \times \sum y \sin\left(\frac{\pi x}{6}\right) = \frac{2}{12} \times 18.36 = 3.06$$

$$a_2 = \frac{2}{n} \times \sum y \cos\left(\frac{2\pi x}{6}\right) = \frac{2}{12} \times (-0.35) = -0.06$$

$$b_2 = \frac{2}{n} \times \sum y \sin\left(\frac{2\pi x}{6}\right) = \frac{2}{12} \times 0.59 = 0.10$$

$$f(x) = \frac{23.88}{2} + 5.9 \cos\left(\frac{\pi x}{6}\right) + 3.06 \sin\left(\frac{\pi x}{6}\right) - 0.06 \cos\left(\frac{2\pi x}{6}\right) + 0.10 \sin\left(\frac{2\pi x}{6}\right)$$

$$f(x) = 11.94 + 5.9 \cos\left(\frac{\pi x}{6}\right) + 3.06 \sin\left(\frac{\pi x}{6}\right) - 0.06 \cos\left(\frac{2\pi x}{6}\right) + 0.10 \sin\left(\frac{2\pi x}{6}\right)$$

35) Compute the first three harmonic for the function $y = f(x)$ from.

x°	0	30	60	90	120	150	180	210	240	270	300	330
y°	25	40	50.5	57.5	61.5	63.5	63.2	60.2	52.5	44.2	39.8	28.7

Here 1st and last values are not repeated.

$$\therefore n = 12$$

y	x	$y \cos x$	$y \sin x$	$y \cos 2x$	$y \sin 2x$	$y \cos 3x$	$y \sin 3x$
25	0	25	0	25	0	25	0
40	30	34.64	20	20	34.64	0	40
50.5	60	25.25	43.73	-25.25	43.73	-50.5	0
57.5	90	0	57.5	-57.5	0	0	-57.5
61.5	120	-30.75	53.26	-30.75	-53.26	61.5	0
63.5	150	-54.99	31.75	31.75	-54.99	0	63.5
63.2	180	-63.2	0	63.2	0	-63.2	0
60.2	210	-52.13	-30.1	30.1	52.13	0	-60.2
52.5	240	-26.25	-45.46	-26.25	45.46	52.5	0
44.2	270	0	-44.2	-44.2	0	0	44.2
39.8	300	19.9	-34.46	-19.9	-34.46	-39.8	0
28.7	330	24.85	-14.35	14.35	-24.85	0	-28.7
$\Sigma y =$ 586.6	-	$\Sigma y \cos x$ -97.68	$\Sigma y \sin x$ 37.67	$\Sigma y \cos 2x$ -19.45	$\Sigma y \sin 2x$ 8.4	$\Sigma y \cos 3x$ -14.5	$\Sigma y \sin 3x$ 1.3

$$a_0 = \frac{2}{n} \times \Sigma y = \frac{2}{12} \times 586.6 = 97.77$$

$$a_1 = \frac{2}{n} \times \Sigma y \cos x = \frac{2}{12} \times (-97.68) = -16.28$$

$$b_1 = \frac{2}{n} \times \Sigma y \sin x = \frac{2}{12} \times (37.67) = 6.28$$

$$a_2 = \frac{2}{n} \times \Sigma y \cos 2x = \frac{2}{12} \times (-19.45) = -3.24$$

$$b_2 = \frac{2}{n} \times \Sigma y \sin 2x = \frac{2}{12} \times (8.4) = 1.4$$

$$a_3 = \frac{2}{n} \times \Sigma y \cos 3x = \frac{2}{12} \times (-14.5) = -2.42$$

$$b_3 = \frac{2}{n} \times \Sigma y \sin 3x = \frac{2}{12} \times (1.3) = 0.22$$

$$f(x) = \frac{a_0}{2} + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + a_3 \cos 3x + b_3 \sin 3x$$

$$= \frac{97.77}{2} - 16.28 \cos x + 6.28 \sin x$$

$$- 3.24 \cos 2x + 1.4 \sin 2x$$

$$- 2.42 \cos 3x + 0.22 \sin 3x$$

$$= 48.89 - 16.28 \cos x + 6.28 \sin x$$

$$- 3.24 \cos 2x + 1.4 \sin 2x$$

$$- 2.42 \cos 3x + 0.22 \sin 3x$$

3b) Evaluate first two harmonics from the following data.

t(sec)	0	T/6	T/3	T/2	2T/3	5T/6	T
A(amp)	1.8	1.1	0.3	0.16	0.5	1.5	1.8

Here 1st and last values are repeated.

$$n = n - 1 = 7 - 1 = 6$$

A (y)	t T = 360°(x)	y cos x	y sin x	y cos 2x	y sin 2x
1.8	0	1.8	0	1.8	0
1.1	$\frac{T}{6} = 60^\circ$	0.55	0.95	-0.55	0.95
0.3	$\frac{T}{3} = 120^\circ$	-0.15	0.26	-0.15	-0.26
0.16	$\frac{T}{2} = 180^\circ$	-0.16	0	0.16	0
0.5	$\frac{2T}{3} = 240^\circ$	-0.25	-0.43	-0.25	0.43
1.5	$\frac{5T}{6} = 300^\circ$	0.75	-1.30	-0.75	-1.30
1.8 Σy = 5.36	—	Σy cos x = 2.54	Σy sin x = -0.52	Σy cos 2x = 0.26	Σy sin 2x = -0.18

$$a_0 = \frac{2}{n} \times \Sigma y = \frac{2}{6} \times 5.36 = 1.79$$

$$a_1 = \frac{2}{n} \times \sum y \cos x = \frac{2}{6} \times 2.54 = 0.85$$

$$b_1 = \frac{2}{n} \times \sum y \sin x = \frac{2}{6} \times (-0.52) = -0.17$$

$$a_2 = \frac{2}{n} \times \sum y \cos 2x = \frac{2}{6} \times 0.26 = 0.09$$

$$b_2 = \frac{2}{n} \times \sum y \sin 2x = \frac{2}{6} \times (-0.18) = -0.06$$

$$f(x) = \frac{a_0}{2} + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x$$

$$= \frac{1.79}{2} + 0.85 \cos x - 0.17 \sin x + 0.09 \cos 2x - 0.06 \sin 2x$$

$$= 0.895 + 0.85 \cos x - 0.17 \sin x$$

$$+ 0.09 \cos 2x - 0.06 \sin 2x$$